CHAPTER 19

Aldehydes and Ketones: Nucleophilic Addition Reactions



FIGURE 19.1 **Few flowers are more beautiful or more fragrant than roses.** Their perfumed odor is due to several simple organic compounds, including the ketone β-damascenone. (credit: modification of work "Rosa 'Precious Platinum'" by Bernard Spragg/Wikimedia Commons, CC BY 1.0)

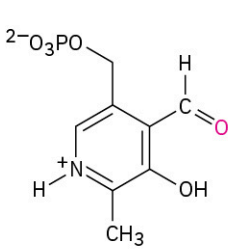
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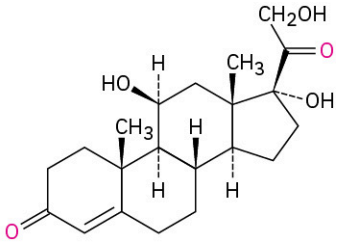
WHY THIS CHAPTER? Much of organic chemistry is the chemistry of carbonyl compounds. Aldehydes and ketones, in particular, are intermediates in the synthesis of many pharmaceutical agents, in almost all biological

pathways, and in numerous industrial processes, so an understanding of their properties and reactions is essential. In this chapter, we'll look at some of their most important reactions.

Aldehydes (RCHO) and **ketones (R_2CO)** are among the most widely occurring of all compounds. In nature, many substances required by living organisms are aldehydes or ketones. The aldehyde pyridoxal phosphate, for instance, is a coenzyme involved in a large number of metabolic reactions; the ketone hydrocortisone is a steroid hormone secreted by the adrenal glands to regulate fat, protein, and carbohydrate metabolism.

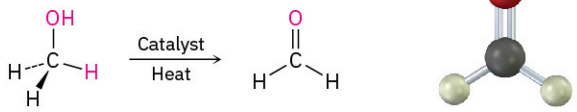


Pyridoxal phosphate (PLP)

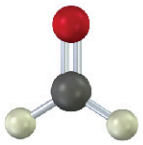


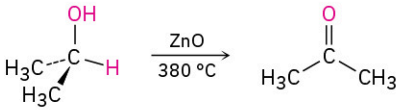
Hydrocortisone

In the chemical industry, simple aldehydes and ketones are produced in large quantities for use as solvents and as starting materials to prepare a host of other compounds. For example, more than 50 million tons per year of formaldehyde, $H_2C{=}O$, is produced worldwide for use in building insulation materials and in the adhesive resins that bind particle board and plywood. Acetone, $(CH_3)_2C{=}O$, is widely used as an industrial solvent, with approximately 8 million tons per year produced worldwide. Formaldehyde is synthesized industrially by catalytic oxidation of methanol, and one method of acetone preparation involves oxidation of 2-propanol produced by hydration of propylene.

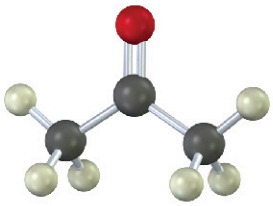


Methanol **Formaldehyde**





2-Propanol **Acetone**

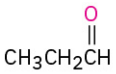


19.1 Naming Aldehydes and Ketones

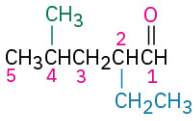
Aldehydes are named by replacing the terminal *-e* of the corresponding alkane name with *-al.* The parent chain must contain the –CHO group, and the –CHO carbon is numbered as carbon 1. Note in the following examples that the longest chain in 2-ethyl-4-methylpentanal is actually a hexane, but this chain does not include the –CHO group and thus is not the parent.



Ethanal (acetaldehyde)

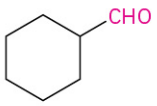


Propanal (propionaldehyde)

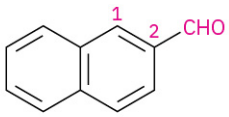


2-Ethyl-4-methylpentanal

For cyclic aldehydes in which the –CHO group is directly attached to a ring, the ring name followed by *carbaldehyde* is used.



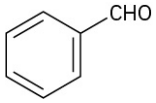
Cyclohexanecarbaldehyde



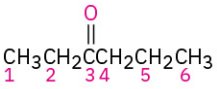
2-Naphthalenecarbaldehyde

A few simple and well-known aldehydes have common names that are recognized by IUPAC. Several that you might encounter are listed in TABLE 19.1.

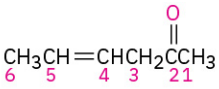
TABLE 19.1 Common Names of Some Simple Aldehydes

Formula	Common name	Systematic name
HCHO	Formaldehyde	Methanal
CH_3CHO	Acetaldehyde	Ethanal
$H_2C{=}CHCHO$	Acrolein	Propenal
$CH_3CH{=}CHCHO$	Crotonaldehyde	2-Butenal
	Benzaldehyde	Benzenecarbaldehyde

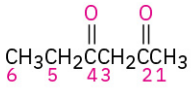
Ketones are named by replacing the terminal *-e* of the corresponding alkane name with *-one*. The parent chain is the longest one that includes the ketone group, and the numbering begins at the end nearer the carbonyl carbon. As with alkenes (Section 7.3) and alcohols (Section 17.1), the locant is placed before the parent name using older rules but before the suffix with the newer IUPAC guidelines. For example:



3-Hexanone
(New: Hexan-3-one)

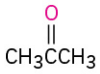


4-Hexen-2-one
(New: Hex-4-en-2-one)

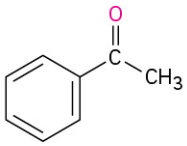


2,4-Hexanedione
(New: Hexane-2,4-dione)

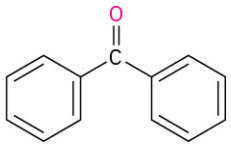
A few ketones are allowed by IUPAC to retain their common names.



Acetone

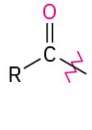


Acetophenone

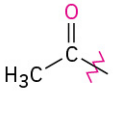


Benzophenone

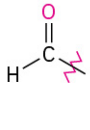
When it's necessary to refer to the R–C=O group as a substituent, the name **acyl (a-sil) group** is used and the name ending *-yl* is attached. Thus, $-COCH_3$ is an *acetyl* group, –CHO is a *formyl* group, –COAr is an *aroyl* group, and $-COC_6H_5$ is a *benzoyl* group.



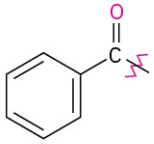
An acyl group



Acetyl

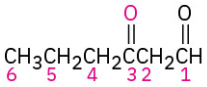


Formyl



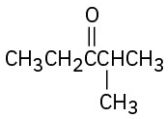
Benzoyl

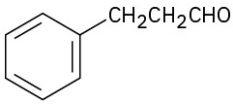
If other functional groups are present and the double-bonded oxygen is considered a substituent on a parent chain, the prefix *oxo-* is used. For example:

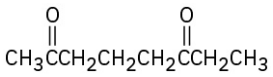


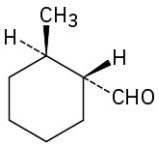
3-Oxohexanal

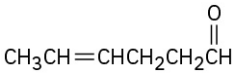
PROBLEM 19-1 Name the following aldehydes and ketones:

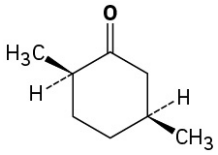
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

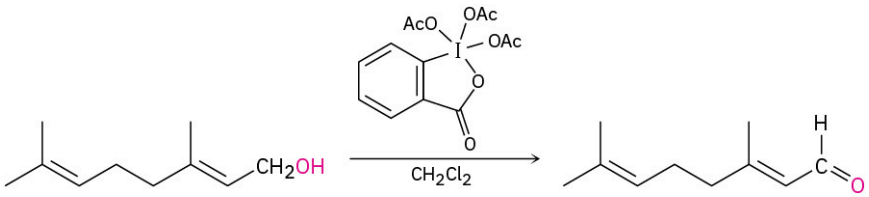
PROBLEM 19-2 Draw structures corresponding to the following names:

(a) 3-Methylbutanal **(b)** 4-Chloro-2-pentanone **(c)** Phenylacetaldehyde **(d)** *cis*-3-*tert*-Butylcyclohexanecarbaldehyde **(e)** 3-Methyl-3-butenal **(f)** 2-(1-Chloroethyl)-5-methylheptanal

19.2 Preparing Aldehydes and Ketones

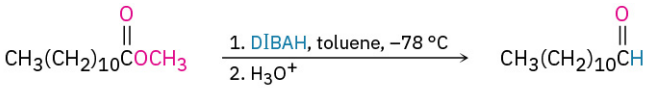
Preparing Aldehydes

Perhaps the best method of aldehyde synthesis is by oxidation of a primary alcohol, as we saw in **Section 17.7**. The reaction is usually carried out using the Dess–Martin periodinane reagent in dichloromethane solvent at room temperature:

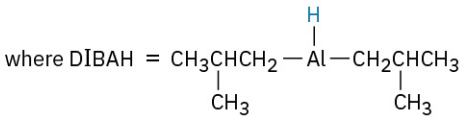


Geraniol **Geranial (84%)**

Another method of aldehyde synthesis is one that we'll mention here just briefly and then return to in **Section 21.6**. Certain carboxylic acid derivatives can be *partially* reduced to yield aldehydes. The partial reduction of an ester by diisobutylaluminum hydride (DIBAH, or DIBAL-H), for instance, is an important laboratory-scale method of aldehyde synthesis, and mechanistically related processes also occur in biological pathways. The reaction is normally carried out at −78 °C (dry-ice temperature) in toluene solution.



Methyl dodecanoate **Dodecanal (88%)**



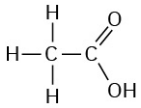
PROBLEM 19-3 How would you prepare pentanal from the following starting materials?

(a) $CH_3CH_2CH_2CH_2CH_2OH$ **(b)** $CH_3CH_2CH_2CH_2CH{=}CH_2$ **(c)** $CH_3CH_2CH_2CH_2CO_2CH_3$ **(d)** $CH_3CH_2CH_2CH{=}CH_2$

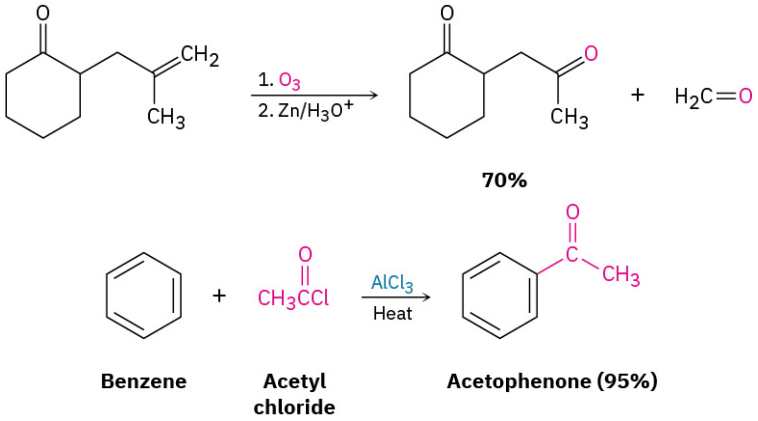
Preparing Ketones

For the most part, methods of ketone synthesis are similar to those for aldehydes. Secondary alcohols are

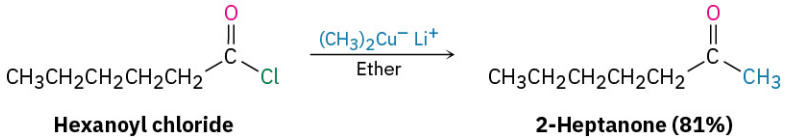
oxidized by a variety of reagents to give ketones (Section 17.7). The choice of oxidant depends on such factors as reaction scale, cost, and acid or base sensitivity of the alcohol. The Dess–Martin periodinane is a common choice although chromic acid (H_2CrO_4) was frequently used in older work.



Other methods include the ozonolysis of alkenes in which one of the unsaturated carbon atoms is disubstituted (Section 8.8) and Friedel–Crafts acylation of an aromatic ring with an acid chloride in the presence of $AlCl_3$ catalyst (Section 16.3).



In addition to those methods already discussed, ketones can also be prepared from certain carboxylic acid derivatives, just as aldehydes can. Among the most useful reactions of this sort is that between an acid chloride and a lithium diorganocopper reagent, as we saw in Section 10.7. We'll discuss this reaction in more detail in **Section 21.4**.

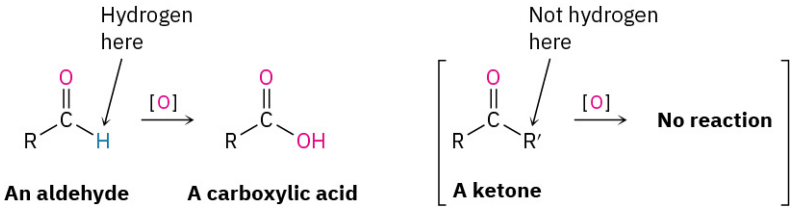


PROBLEM How would you carry out the following reactions? More than one step may be needed.

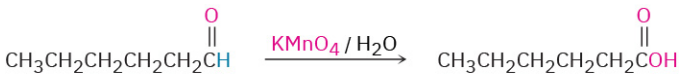
19-4 **(a)** 3-Hexyne → 3-Hexanone **(b)** Benzene → m-Bromoacetophenone
(c) Bromobenzene → Acetophenone **(d)** 1-Methylcyclohexene → 2-Methylcyclohexanone

19.3 Oxidation of Aldehydes and Ketones

Aldehydes are easily oxidized to yield carboxylic acids, but ketones are generally inert toward oxidation. The difference is a consequence of structure: aldehydes have a –C**H**O proton that can be abstracted during oxidation, but ketones do not.

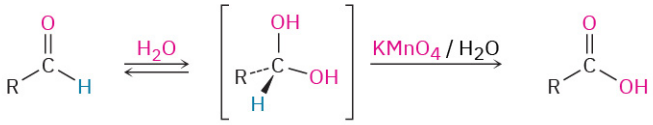


Many oxidizing agents, including alkaline $KMnO_4$ and hot HNO_3, convert aldehydes into carboxylic acids. The oxidation occurs rapidly at room temperature and generally works well.



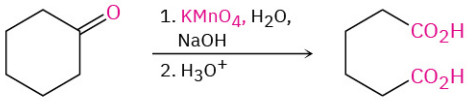
Hexanal **Hexanoic acid (85%)**

Aldehyde oxidations occur through intermediate 1,1-diols, or *hydrates,* which are formed by a reversible nucleophilic addition of water to the carbonyl group. Even though it's formed to only a small extent at equilibrium, the hydrate reacts like any typical primary or secondary alcohol and is oxidized to a carbonyl compound (Section 17.7).



An aldehyde **A hydrate** **A carboxylic acid**

Ketones are inert to most oxidizing agents but undergo a slow cleavage reaction of the C–C bond next to the carbonyl group when treated with hot alkaline $KMnO_4$. The reaction is not often used, however, and is mentioned here only for completeness.



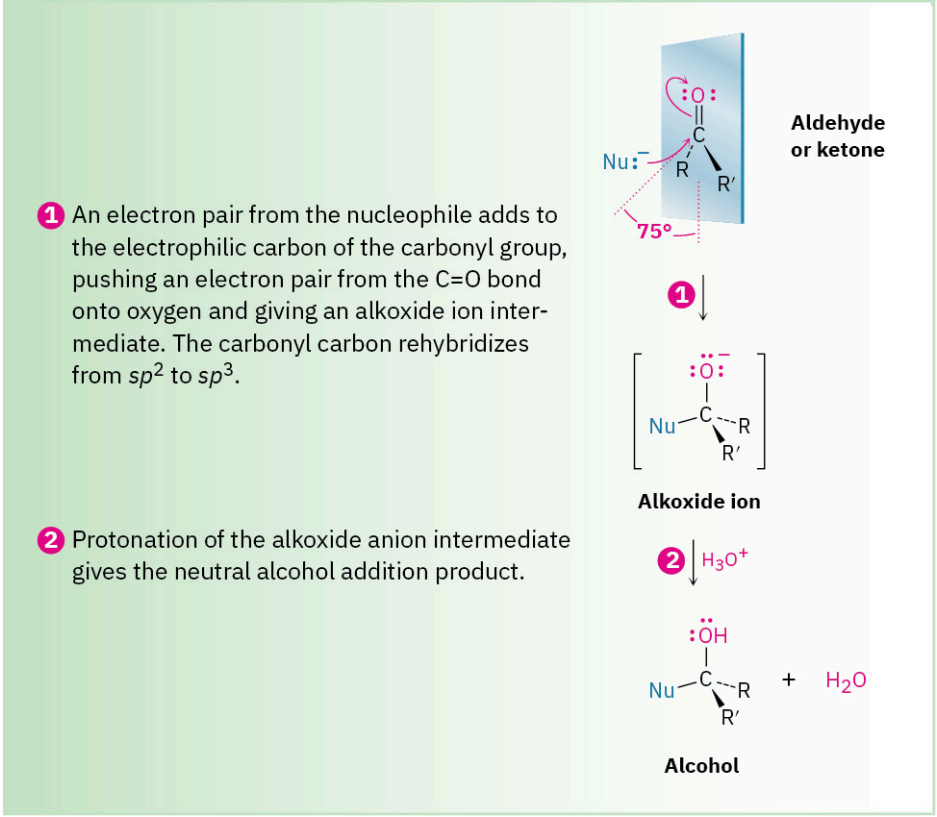
Cyclohexanone **Hexanedioic acid (79%)**

19.4 Nucleophilic Addition Reactions of Aldehydes and Ketones

As we saw in the ***Preview of Carbonyl Chemistry,*** the most general reaction of aldehydes and ketones is the **nucleophilic addition reaction**. As shown in FIGURE 19.2, a nucleophile, **:**Nu^-, approaches the carbonyl group from an angle of about 105° opposite the carbonyl oxygen and forms a bond to the electrophilic C–O carbon atom. At the same time, rehybridization of the carbonyl carbon from sp^2 to sp^3 occurs, an electron pair from the C═O bond moves toward the electronegative oxygen atom, and a tetrahedral alkoxide ion intermediate is produced. Protonation of the alkoxide by addition of acid then gives an alcohol.

FIGURE 19.2 MECHANISM

A nucleophilic addition reaction to an aldehyde or ketone. The nucleophile approaches the carbonyl group from an angle of approximately 75° to the plane of the sp^2 orbitals, the carbonyl carbon rehybridizes from sp^2 to sp^3, and an alkoxide ion is formed. Protonation by addition of acid then gives an alcohol.



The nucleophile can be either negatively charged (:Nu$^-$) or neutral (:Nu). If it's neutral, however, it usually carries a hydrogen atom that can subsequently be eliminated, :Nu–H. For example:

Some negatively charged nucleophiles

- $HO:^-$ (hydroxide ion)
- $H:^-$ (hydride ion)
- $R_3C:^-$ (a carbanion)
- $RO:^-$ (an alkoxide ion)
- $N{\equiv}C:^-$ (cyanide ion)

Some neutral nucleophiles

- HOH (water)
- ROH (an alcohol)
- H_3N: (ammonia)
- RNH_2 (an amine)

Nucleophilic additions to aldehydes and ketones have two general variations, as shown in FIGURE 19.3. In one variation, the tetrahedral intermediate is protonated by water or acid to give an alcohol as the final product. In the second variation, the carbonyl oxygen atom is protonated and then eliminated as HO^- or H_2O to give a product with a C═Nu double bond.

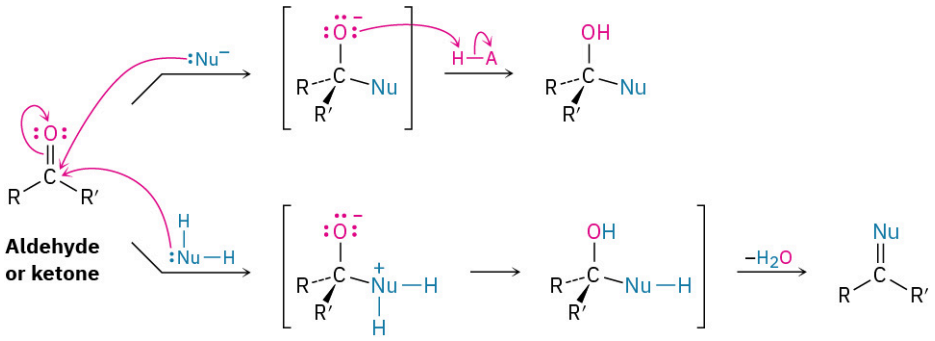


FIGURE 19.3 Two general reaction pathways following addition of a nucleophile to an aldehyde or ketone. The top pathway leads to an alcohol product; the bottom pathway leads to a product with a C=Nu double bond.

Aldehydes are generally more reactive than ketones in nucleophilic addition reactions for both steric and electronic reasons. Sterically, the presence of only one large substituent bonded to the C=O carbon in an aldehyde versus two large substituents in a ketone means that a nucleophile is able to approach an aldehyde more readily. Thus, the transition state leading to the tetrahedral intermediate is less crowded and lower in energy for an aldehyde than for a ketone (**FIGURE 19.4**).

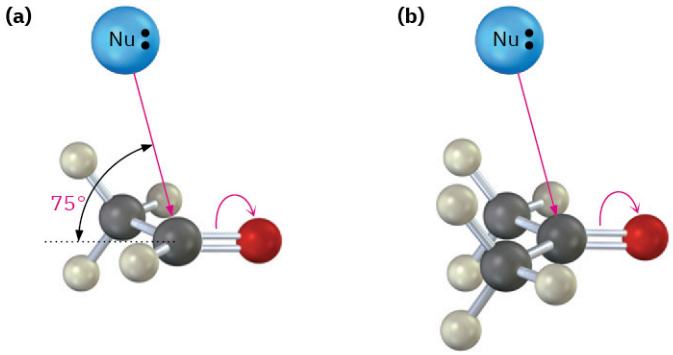


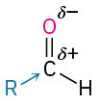
FIGURE 19.4 Steric hindrance in nucleophilic addition reactions. (a) Nucleophilic addition to an aldehyde is sterically less hindered because only one relatively large substituent is attached to the carbonyl-group carbon. **(b)** A ketone, however, has two large substituents and is more hindered. The approach of the nucleophile is along the C=O bond at an angle of about 75° to the plane of the carbon sp^2 orbitals.

Electronically, aldehydes are more reactive than ketones because of the greater polarization of aldehyde carbonyl groups. To see this polarity difference, recall the stability order of carbocations (**Section 7.9**). A primary carbocation is higher in energy and thus more reactive than a secondary carbocation since it has only one alkyl group inductively stabilizing the positive charge rather than two. In the same way, an aldehyde has only one alkyl group inductively stabilizing the partial positive charge on the carbonyl carbon rather than two, and is a bit more electrophilic, and, therefore, more reactive than a ketone.

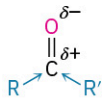


1° carbocation
(less stable, more reactive)

2° carbocation
(more stable, less reactive)



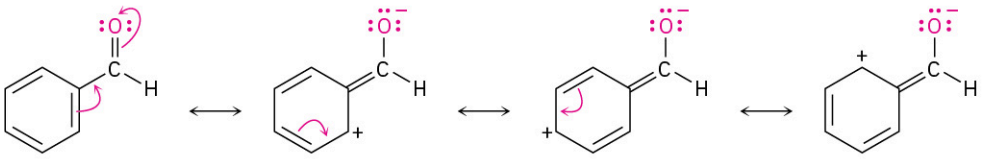
Aldehyde
(less stabilization of δ+, more reactive)

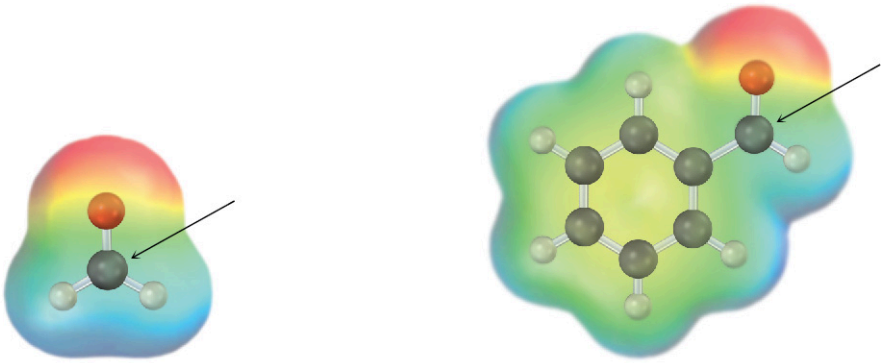


Ketone
(more stabilization of δ+, less reactive)

One further comparison: aromatic aldehydes, such as benzaldehyde, are less reactive in nucleophilic addition reactions than aliphatic aldehydes because the electron-donating resonance effect of the aromatic ring makes the carbonyl group less electrophilic. Comparing electrostatic potential maps of formaldehyde and benzaldehyde, for example, shows that the carbonyl carbon atom is less positive (less blue) in the aromatic

aldehyde.





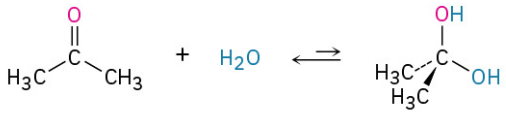
Formaldehyde **Benzaldehyde**

PROBLEM 19-5 Treatment of an aldehyde or ketone with cyanide ion ($^{-}$:C≡N), followed by protonation of the tetrahedral alkoxide ion intermediate, gives a *cyanohydrin.* Show the structure of the cyanohydrin obtained from cyclohexanone.

PROBLEM 19-6 *p*-Nitrobenzaldehyde is more reactive toward nucleophilic additions than *p*-methoxybenzaldehyde. Explain.

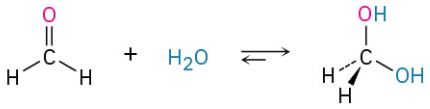
19.5 Nucleophilic Addition of H_2O: Hydration

Aldehydes and ketones react with water to yield 1,1-diols, or *geminal (gem)* diols. The hydration reaction is reversible, and a gem diol can eliminate water to regenerate an aldehyde or ketone.



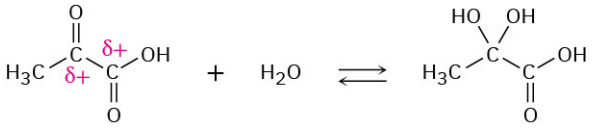
Acetone (99.9%) **Acetone hydrate (0.1%)**

The position of the equilibrium between a gem diol and an aldehyde or ketone depends on the structure of the carbonyl compound. Equilibrium generally favors the carbonyl compound for steric reasons, but the gem diol is favored for a few simple aldehydes. For example, an aqueous solution of formaldehyde consists of 99.9% gem diol and 0.1% aldehyde at equilibrium, whereas an aqueous solution of acetone consists of only about 0.1% gem diol and 99.9% ketone.

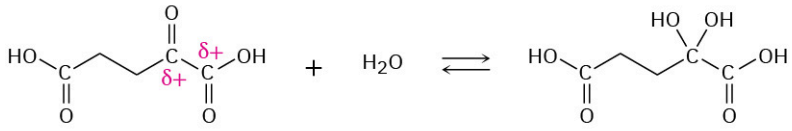


Formaldehyde (0.1%) **Formaldehyde hydrate (99.9%)**

The hydrate is also favored for α-keto carboxylic acids, which have adjacent positively charged carbons that destabilize the keto form and favor the hydrate. Such compounds occur commonly in many biological pathways. Pyruvic acid, which is 60% hydrate at equilibrium, and α-ketoglutaric acid, which is 50% hydrate, are examples.



Pyruvic acid (40%) **Pyruvic acid hydrate (60%)**



α-Ketoglutaric acid (50%) **α-Ketoglutaric acid hydrate (50%)**

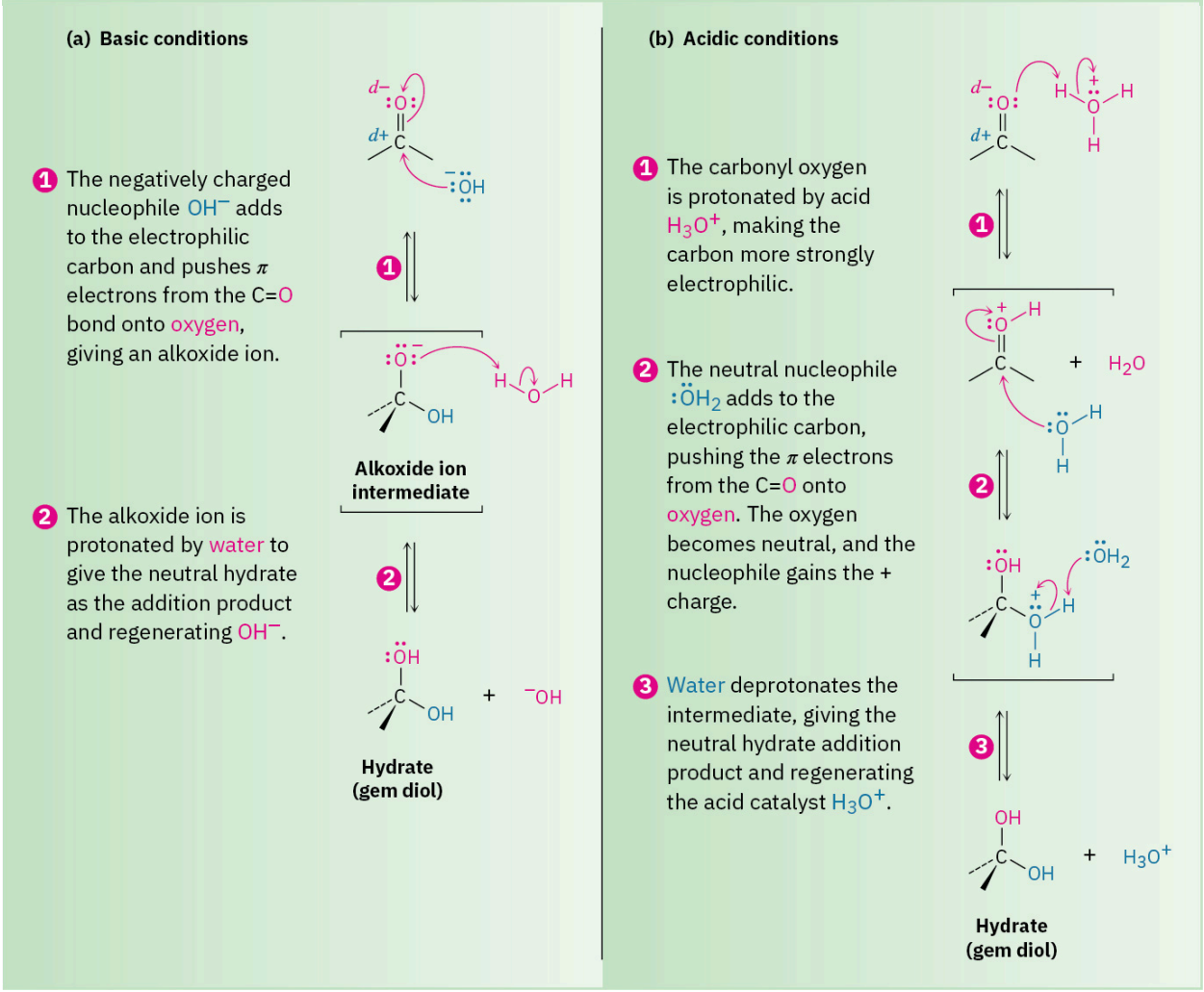
The nucleophilic addition of water to an aldehyde or ketone is slow under neutral conditions but is catalyzed by both base and acid. Under basic conditions (**FIGURE 19.5**a), the nucleophile is negatively charged (OH^-) and uses a pair of its electrons to form a bond to the electrophilic carbon atom of the C═O group. At the same time, the C═O carbon atom rehybridizes from sp^2 to sp^3 and two electrons from the C═O π bond are pushed onto the oxygen atom, giving an alkoxide ion. Protonation of the alkoxide ion by water then yields a neutral addition product plus regenerated OH^-.

Under acidic conditions (**FIGURE 19.5**b), the carbonyl oxygen atom is first protonated by H_3O^+ to make the carbonyl group more strongly electrophilic. A neutral nucleophile, H_2O, then uses a pair of electrons to bond to the carbon atom of the C═O group, and two electrons from the C═O π bond move onto the oxygen atom. The positive charge on oxygen is thereby neutralized, while the nucleophile gains a positive charge. Finally, deprotonation by water gives the neutral addition product and regenerates the H_3O^+ catalyst.

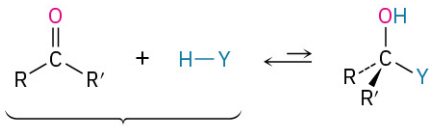
Note the key difference between the base-catalyzed and acid-catalyzed reactions. The base-catalyzed reaction takes place rapidly because water is converted into hydroxide ion, a much better *nucleophile.* The acid-catalyzed reaction takes place rapidly because the carbonyl compound is converted by protonation into a much better *electrophile.*

FIGURE 19.5 MECHANISM

The mechanism for a nucleophilic addition reaction of aldehydes and ketones under both basic and acidic conditions. (a) Under basic conditions, a negatively charged nucleophile adds to the carbonyl group to give an alkoxide ion intermediate, which is subsequently protonated. **(b)** Under acidic conditions, protonation of the carbonyl group occurs first, followed by addition of a neutral nucleophile and subsequent deprotonation.



The hydration reaction just described is typical of what happens when an aldehyde or ketone is treated with a nucleophile of the type H–Y, where the Y atom is electronegative and can stabilize a negative charge (oxygen, halogen, or sulfur, for instance). In such reactions, the nucleophilic addition is reversible, with the equilibrium generally favoring the carbonyl reactant rather than the tetrahedral addition product. In other words, treatment of an aldehyde or ketone with CH_3OH, H_2O, HCl, HBr, or H_2SO_4 does not normally lead to a stable alcohol addition product.



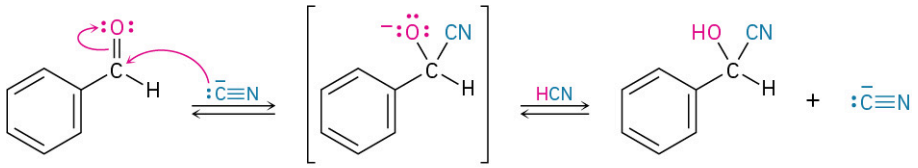
PROBLEM 19-7 When dissolved in water, trichloroacetaldehyde exists primarily as its hydrate, called chloral hydrate. Show the structure of chloral hydrate.

PROBLEM 19-8 The oxygen in water is primarily (99.8%) ^{16}O, but water enriched with the heavy isotope ^{18}O is also available. When an aldehyde or ketone is dissolved in ^{18}O-enriched water, the isotopic label becomes incorporated into the carbonyl group. Explain.

$$R_2C{=}O + H_2O \rightleftharpoons R_2C{=}O + H_2O \quad \text{where O} = {}^{18}O$$

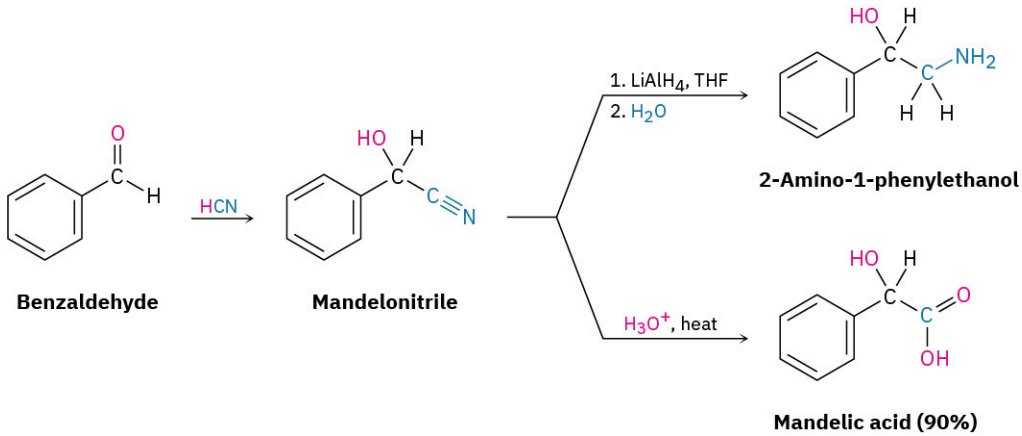
19.6 Nucleophilic Addition of HCN: Cyanohydrin Formation

Aldehydes and unhindered ketones undergo a nucleophilic addition reaction with HCN to yield **cyanohydrins**, RCH(OH)C≡N. Studies carried out in the early 1900s by Arthur Lapworth showed that cyanohydrin formation is reversible and base-catalyzed. Reaction occurs slowly when pure HCN is used but rapidly when a small amount of base is added to generate the nucleophilic cyanide ion, CN^-. Addition of CN^- takes place by a typical nucleophilic addition pathway, yielding a tetrahedral intermediate that is protonated by HCN to give cyanohydrin product plus regenerated CN^-.



Cyanohydrin formation is somewhat unusual because it is one of the few examples of the addition of a protic acid (H–Y) to a carbonyl group. As noted in the previous section, protic acids such as H_2O, HBr, HCl, and H_2SO_4 don't normally yield carbonyl addition products because the equilibrium constants are unfavorable. With HCN, however, equilibrium favors the cyanohydrin adduct.

Cyanohydrin formation is useful because of the further chemistry that can be carried out on the product. For example, a nitrile (R–C≡N) can be reduced with $LiAlH_4$ to yield a primary amine (RCH_2NH_2) and can be hydrolyzed by hot aqueous acid to yield a carboxylic acid. Thus, cyanohydrin formation provides a method for transforming an aldehyde or ketone into a different functional group.



PROBLEM 19-9 Cyclohexanone forms a cyanohydrin in good yield but 2,2,6-trimethylcyclohexanone does not. Explain.

19.7 Nucleophilic Addition of Hydride and Grignard Reagents: Alcohol Formation

Addition of Hydride Reagents: Reduction

We saw in **Section 17.4** that the most common method for preparing alcohols, both in the laboratory and in living organisms, is by the reduction of carbonyl compounds. Aldehydes are reduced with sodium borohydride ($NaBH_4$) to give primary alcohols, and ketones are reduced similarly to give secondary alcohols.



Aldehyde **1° Alcohol** **Ketone** **2° Alcohol**

Carbonyl reduction occurs by a typical nucleophilic addition mechanism under basic conditions, as shown earlier in FIGURE 19.5a. Although the details of carbonyl-group reductions are complex, $LiAlH_4$ and $NaBH_4$ act as if they were donors of hydride ion nucleophile, $:H^-$, and the initially formed alkoxide ion intermediate is then protonated by addition of aqueous acid. The reaction is effectively irreversible because the reverse process would require expulsion of a very poor leaving group.

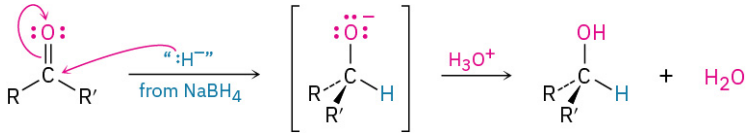


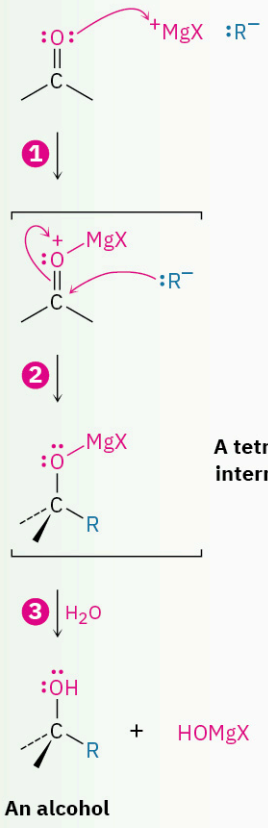
FIGURE 19.6 MECHANISM

Mechanism of the Grignard reaction. Complexation of the carbonyl oxygen with the Lewis acid Mg^{2+} and subsequent nucleophilic addition of a carbanion to an aldehyde or ketone is followed by protonation of the alkoxide intermediate to yield an alcohol.

1. The Lewis acid Mg^{2+} first forms an acid–base complex with the basic oxygen atom of the aldehyde or ketone, thereby making the carbonyl group a better acceptor.

2. Nucleophilic addition of an alkyl group $:R^-$ to the aldehyde or ketone produces a tetrahedral magnesium alkoxide intermediate . . .

3. . . . which undergoes hydrolysis when water is added in a separate step. The final product is a neutral alcohol.



An alcohol

Addition of Grignard Reagents, RMgX

Just as aldehydes and ketones undergo nucleophilic addition with hydride ion to give alcohols, they undergo a similar addition with Grignard reagent nucleophiles, $R:^- {}^+MgX$. Aldehydes give secondary alcohols on reaction with Grignard reagents in ether solution, and ketones give tertiary alcohols.

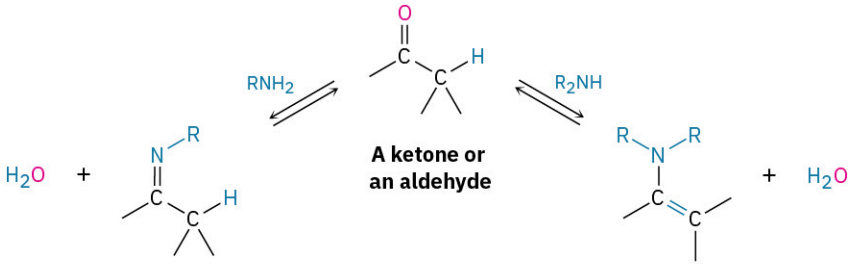


Aldehyde **2° Alcohol** **Ketone** **3° Alcohol**

As shown in **FIGURE 19.6**, a Grignard reaction begins with an acid–base complexation of Mg^{2+} to the carbonyl oxygen atom of the aldehyde or ketone, thereby making the carbonyl group a better electrophile. Nucleophilic addition of $R:^-$ then produces a tetrahedral magnesium alkoxide intermediate, and protonation by addition of water or dilute aqueous acid in a separate step yields the neutral alcohol. Like reduction, Grignard additions are effectively irreversible because a carbanion is too poor a leaving group to be expelled in a reversal step.

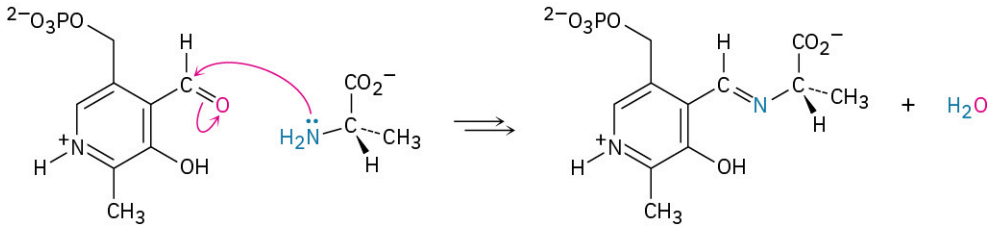
19.8 Nucleophilic Addition of Amines: Imine and Enamine Formation

Primary amines, RNH_2, add to aldehydes and ketones to yield **imines**, $R_2C{=}NR$. Secondary amines, R_2NH, add similarly to yield **enamines**, $R_2N{-}CR{=}CR_2$ (*ene* + *amine* = unsaturated amine).



An imine **An enamine**

Imines are particularly common as intermediates in biological pathways, where they are often called **Schiff bases**. The amino acid alanine, for instance, is metabolized in the body by reaction with the aldehyde pyridoxal phosphate (PLP), a derivative of vitamin B_6, to yield a Schiff base that is further degraded.



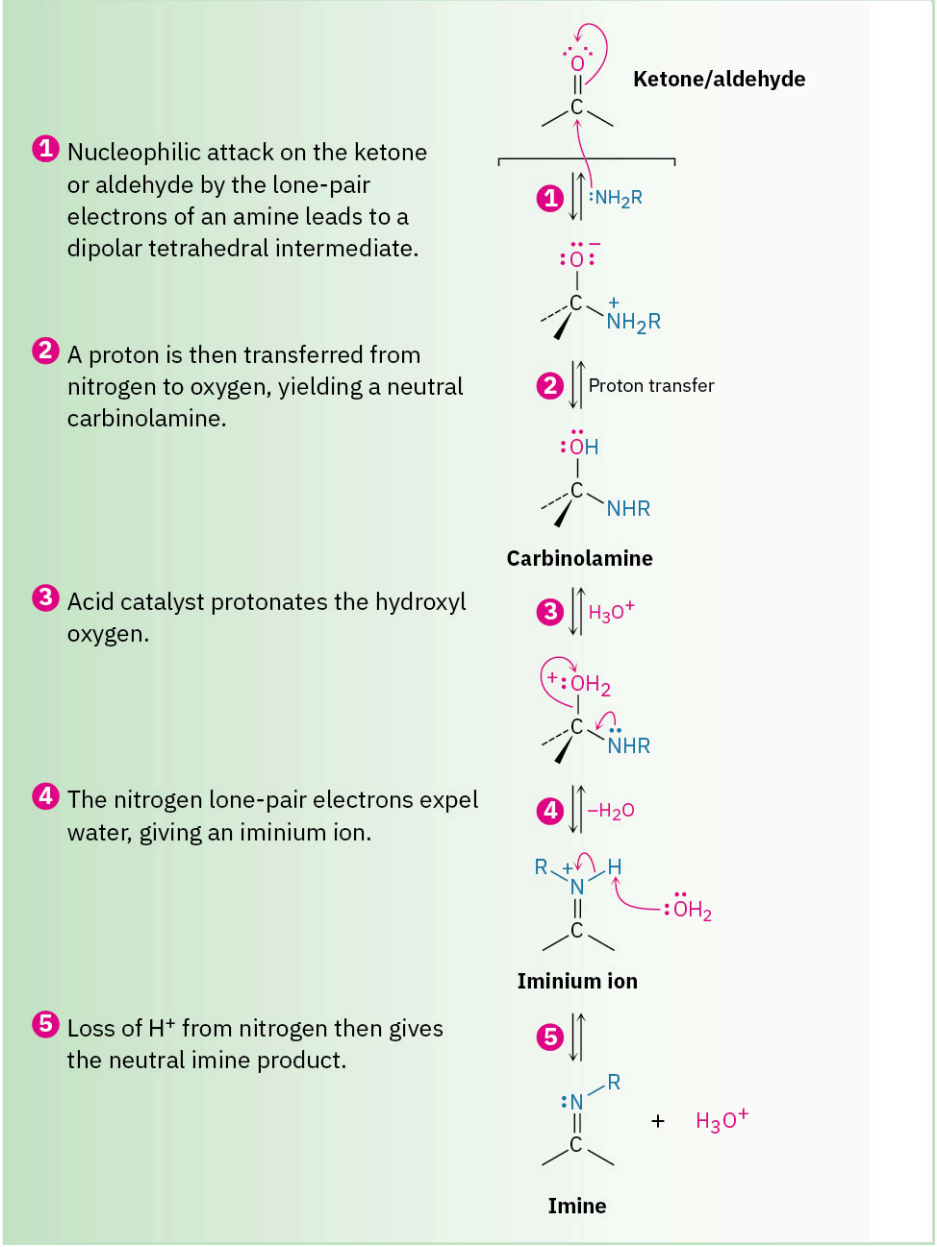
Pyridoxal phosphate **Alanine** **An imine (Schiff base)**

Imine formation and enamine formation seem different because one leads to a product with a C=N bond and the other leads to a product with a C=C bond. Actually, though, the reactions are quite similar. Both are typical examples of nucleophilic addition reactions in which water is eliminated from the initially formed tetrahedral intermediate and a new C=Nu double bond is formed.

Imines are formed in a reversible, acid-catalyzed process (**FIGURE 19.7**) that begins with nucleophilic addition of the primary amine to the carbonyl group, followed by transfer of a proton from nitrogen to oxygen to yield a neutral amino alcohol, or **carbinolamine**. Protonation of the carbinolamine oxygen by an acid catalyst then converts the –OH into a better leaving group ($-OH_2^+$), and E1-like loss of water produces an iminium ion. Loss of a proton from nitrogen gives the final product and regenerates the acid catalyst.

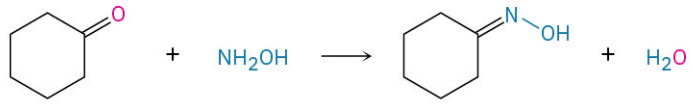
FIGURE 19.7 MECHANISM

Mechanism of imine formation by reaction of an aldehyde or ketone with a primary amine. The key step is the initial nucleophilic addition to yield a carbinolamine intermediate, which then loses water to give the imine.



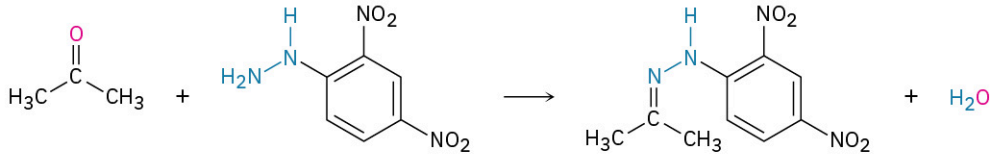
Imine formation with such reagents as hydroxylamine and 2,4-dinitrophenylhydrazine is sometimes useful because the products of these reactions—*oximes* and *2,4-dinitrophenylhydrazones (2,4-DNPs),* respectively—are often crystalline and easy to handle. Such crystalline derivatives are occasionally prepared as a means of purifying and characterizing liquid ketones or aldehydes.

Oxime



Cyclohexanone **Hydroxylamine** **Cyclohexanone oxime (mp 90 °C)**

2,4-Dinitrophenyl-hydrazone

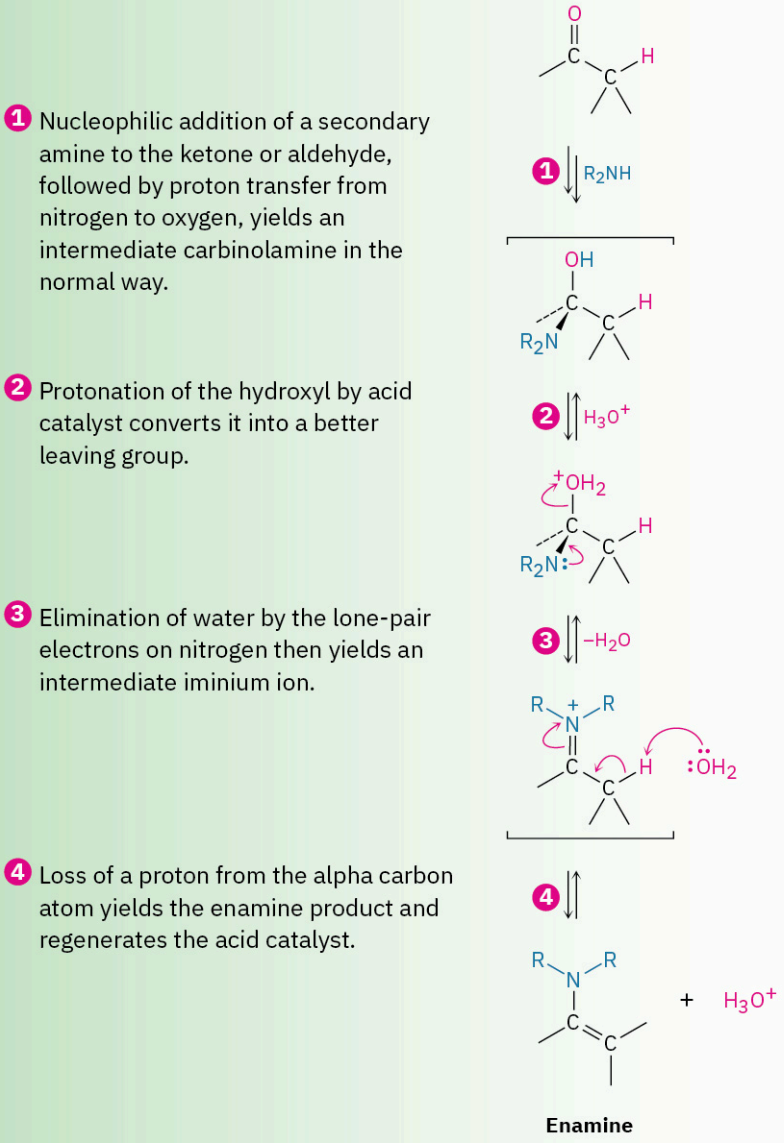


Acetone **2,4-Dinitrophenyl-hydrazine** **Acetone 2,4-dinitrophenyl-hydrazone (mp 126 °C)**

Reaction of an aldehyde or ketone with a secondary amine, R_2NH, rather than a primary amine yields an enamine. As shown in **FIGURE 19.8**, the process is identical to imine formation up to the iminium ion stage, but at this point there is no proton on nitrogen that can be lost to form a neutral imine product. Instead, a proton is lost from the *neighboring* carbon (the α carbon), yielding an enamine.

FIGURE 19.8 MECHANISM

Mechanism for enamine formation by reaction of an aldehyde or ketone with a secondary amine, R_2NH. The iminium ion intermediate formed in step 3 has no hydrogen attached to N and so must lose H^+ from the carbon two atoms away.



Imine and enamine formation are slow at both high pH and low pH but reach a maximum rate at a weakly acidic pH around 4 to 5. For example, the profile of pH versus rate shown in **FIGURE 19.9** for the reaction between acetone and hydroxylamine, NH_2OH, indicates that the maximum reaction rate occurs at pH 4.5.

We can explain the observed pH dependence of imine formation by looking at the individual steps in the mechanism. As indicated in **FIGURE 19.8**, an acid catalyst is required in step 3 to protonate the intermediate carbinolamine, thereby converting the –OH into a better leaving group. Thus, reaction will be slow if not enough acid is present (that is, at high pH). On the other hand, if too much acid is present (low pH), the basic amine nucleophile is completely protonated, so the initial nucleophilic addition step can't occur.

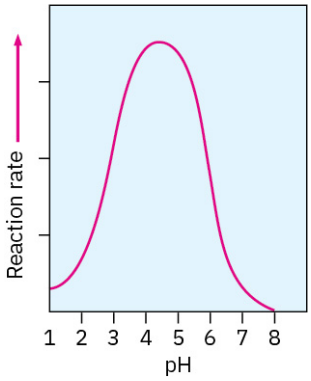


FIGURE 19.9 Dependence on pH of the rate of reaction between acetone and hydroxylamine: $(CH_3)_2C{=}O + NH_2OH \rightarrow (CH_3)_2C{=}NOH + H_2O$.

Evidently, a pH of 4.5 represents a compromise between the need for *some* acid to catalyze the rate-limiting dehydration step but *not too much* acid so as to avoid complete protonation of the amine. Each nucleophilic addition reaction has its own requirements, and conditions must be optimized to obtain maximum reaction rates.

WORKED EXAMPLE 19.1

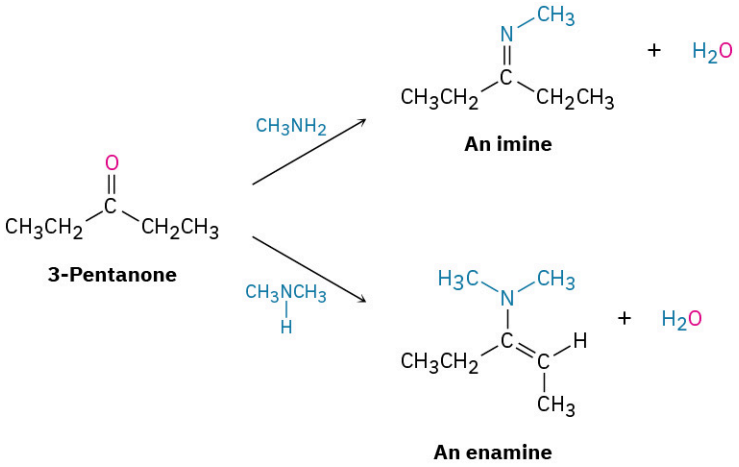
Predicting the Product of Reaction between a Ketone and an Amine

Show the products you would obtain by acid-catalyzed reaction of 3-pentanone with methylamine, CH_3NH_2, and with dimethylamine, $(CH_3)_2NH$.

Strategy

An aldehyde or ketone reacts with a primary amine, RNH_2, to yield an imine, in which the carbonyl oxygen atom has been replaced by the =N–R group of the amine. Reaction of the same aldehyde or ketone with a secondary amine, R_2NH, yields an enamine, in which the oxygen atom has been replaced by the $-NR_2$ group of the amine and the double bond has moved to a position between the former carbonyl carbon and the neighboring carbon.

Solution

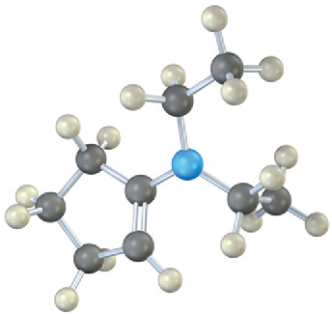


PROBLEM 19-10 Show the products you would obtain by acid-catalyzed reaction of cyclohexanone with ethylamine, $CH_3CH_2NH_2$, and with diethylamine, $(CH_3CH_2)_2NH$.

PROBLEM 19-11 Imine formation is reversible. Show all the steps involved in the acid-catalyzed reaction of an imine with water (hydrolysis) to yield an aldehyde or ketone plus primary amine.

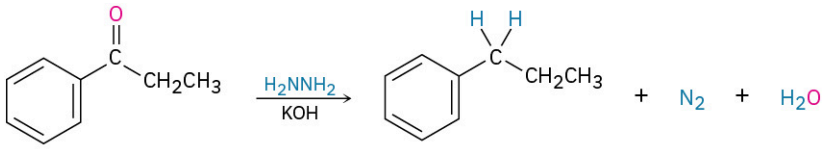
PROBLEM 19-12 Draw the following molecule as a skeletal structure, and show how it can be prepared from a ketone

and an amine.

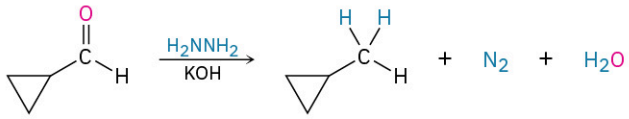


19.9 Nucleophilic Addition of Hydrazine: The Wolff–Kishner Reaction

A useful variant of the imine-forming reaction just discussed involves the treatment of an aldehyde or ketone with hydrazine, $H_2N–NH_2$, in the presence of KOH. Called the **Wolff–Kishner reaction**, the process is a useful and general method for converting an aldehyde or ketone into an alkane, $R_2C{=}O \rightarrow R_2CH_2$.



Propiophenone **Propylbenzene (82%)**

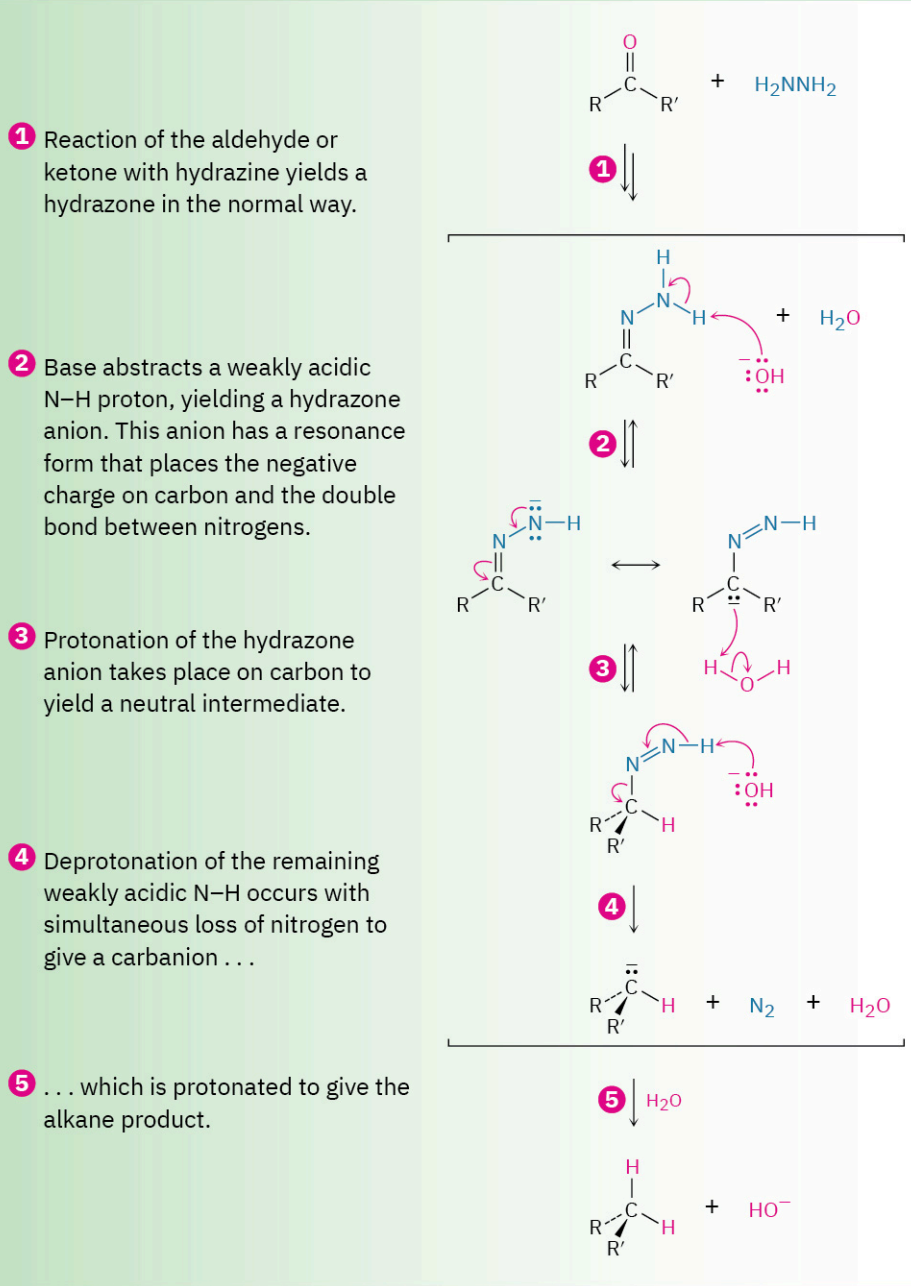


Cyclopropane-carbaldehyde **Methylcyclo-propane (72%)**

As shown in **FIGURE 19.10**, the Wolff–Kishner reaction involves formation of a *hydrazone* intermediate, $R_2C{=}N–NH_2$, followed by base-catalyzed double-bond migration, loss of N_2 gas to give a carbanion, and protonation to give the alkane product. The double-bond migration takes place when a base removes one of the weakly acidic NH protons in step 2 to generate a hydrazone anion, which has an allylic resonance structure that places the double bond between nitrogens and the negative charge on carbon. Reprotonation then occurs on carbon to generate the double-bond rearrangement product. The next step—loss of nitrogen and formation of an alkyl anion—is driven by the large thermodynamic stability of the N_2 molecule.

FIGURE 19.10 MECHANISM

Mechanism for the Wolff–Kishner reduction of an aldehyde or ketone to yield an alkane.



Note that the Wolff–Kishner reduction accomplishes the same overall transformation as the catalytic hydrogenation of an acylbenzene to yield an alkylbenzene (Section 16.10). The Wolff–Kishner reduction is more general and more useful than catalytic hydrogenation, however, because it works well with both alkyl and aryl ketones.

PROBLEM 19-13 Show how you could prepare the following compounds from 4-methyl-3-penten-2-one, $(CH_3)_2C{=}CHCOCH_3$.

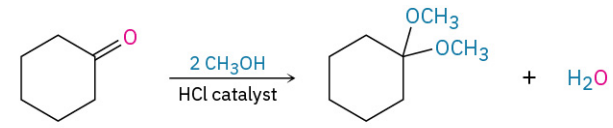
(a) $CH_3CHCH_2CCH_3$ (CH₃ and O)

(b) $CH_3C{=}CHCH_2CH_3$ (CH₃)

(c) $CH_3CHCH_2CH_2CH_3$ (CH₃)

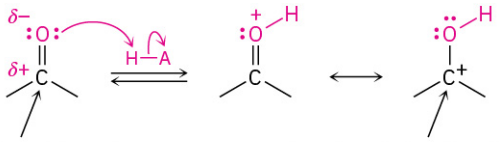
19.10 Nucleophilic Addition of Alcohols: Acetal Formation

Aldehydes and ketones react reversibly with 2 equivalents of an alcohol in the presence of an acid catalyst to yield **acetals, $R_2C(OR')_2$**, which are often called **ketals** if derived from a ketone. Cyclohexanone, for instance, reacts with methanol in the presence of HCl to give the corresponding dimethyl acetal.



Cyclohexanone **Cyclohexanone dimethyl acetal**

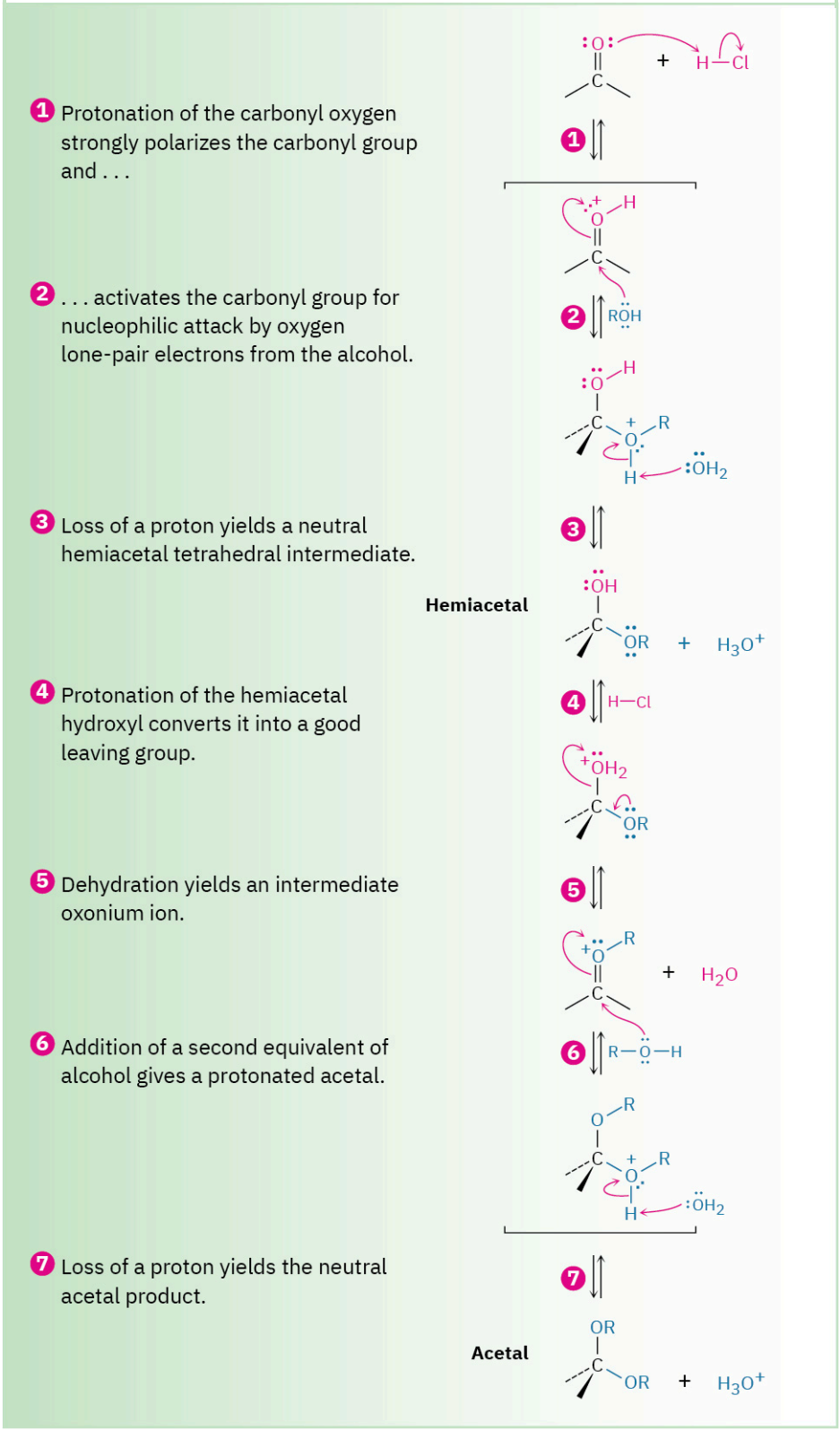
Acetal formation is similar to the hydration reaction discussed in Section 19.5. Like water, alcohols are weak nucleophiles that add to aldehydes and ketones slowly under neutral conditions. Under acidic conditions, however, the reactivity of the carbonyl group is increased by protonation, so addition of an alcohol occurs rapidly.



As shown in FIGURE 19.11, nucleophilic addition of an alcohol to the carbonyl group initially yields a hydroxy ether called a **hemiacetal**, analogous to the gem diol formed by addition of water. Hemiacetals are formed reversibly, with equilibrium normally favoring the carbonyl compound. In the presence of acid, however, a further reaction occurs. Protonation of the –OH group, followed by an E1-like loss of water, leads to an oxonium ion, $R_2C{=}O^+R$, which undergoes a second nucleophilic addition of alcohol to yield the protonated acetal. Loss of a proton completes the reaction.

FIGURE 19.11 MECHANISM

Mechanism of acid-catalyzed acetal formation by reaction of an aldehyde or ketone with an alcohol.



Because all the steps in acetal formation are reversible, the reaction can be driven either forward (from carbonyl

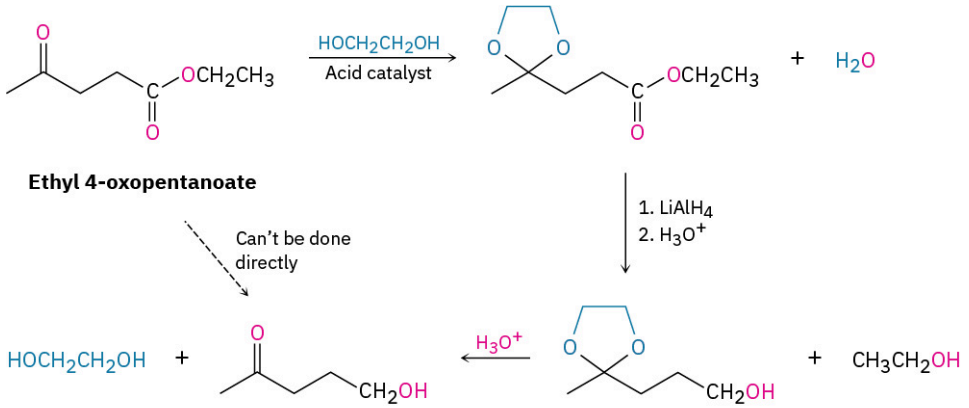
compound to acetal) or backward (from acetal to carbonyl compound), depending on the conditions. The forward reaction is favored by conditions that remove water from the medium and thus drive the equilibrium to the right. In practice, this is often done by distilling off water as it forms. The reverse reaction is favored by treating the acetal with a large excess of aqueous acid to drive the equilibrium to the left.

Acetals are useful because they can act as protecting groups for aldehydes and ketones in the same way that trimethylsilyl ethers act as protecting groups for alcohols (**Section 17.8**). As we saw previously, it sometimes happens that one functional group interferes with intended chemistry elsewhere in a complex molecule. For example, if we wanted to reduce only the ester group of ethyl 4-oxopentanoate, the ketone would interfere. Treatment of the starting keto ester with $LiAlH_4$ would reduce both the keto group and the ester group to give a diol product.

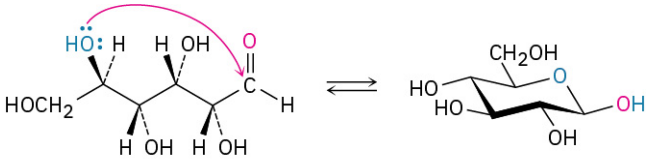


Ethyl 4-oxopentanoate **5-Hydroxy-2-pentanone**

By protecting the keto group as an acetal, however, the problem can be circumvented. Like other ethers, acetals are unreactive to bases, hydride reducing agents, Grignard reagents, and catalytic hydrogenation conditions, but they are cleaved by acid. Thus, we can accomplish the selective reduction of the ester group in ethyl 4-oxopentanoate by first converting the keto group to an acetal, then reducing the ester with $LiAlH_4$, and then removing the acetal by treatment with aqueous acid. (In practice, it's often convenient to use 1 equivalent of a diol such as ethylene glycol as the alcohol to form a *cyclic* acetal. The mechanism of cyclic acetal formation using 1 equivalent of ethylene glycol is exactly the same as that using 2 equivalents of methanol or other monoalcohols.)



Acetal and hemiacetal groups are particularly common in carbohydrate chemistry. Glucose, for instance, is a polyhydroxy aldehyde that undergoes an internal nucleophilic addition reaction and exists primarily as a cyclic hemiacetal.



Glucose—open chain **Glucose—cyclic hemiacetal**

WORKED EXAMPLE 19.2

Predicting the Product of Reaction between a Ketone and an Alcohol

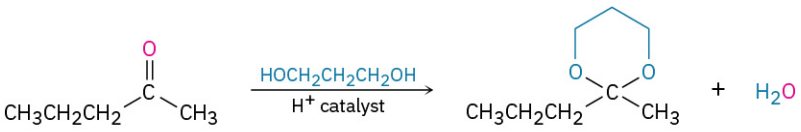
Show the structure of the acetal you would obtain by acid-catalyzed reaction of 2-pentanone with

1,3-propanediol.

Strategy

Acid-catalyzed reaction of an aldehyde or ketone with 2 equivalents of a monoalcohol or 1 equivalent of a diol yields an acetal, in which the carbonyl oxygen atom is replaced by two –OR groups from the alcohol.

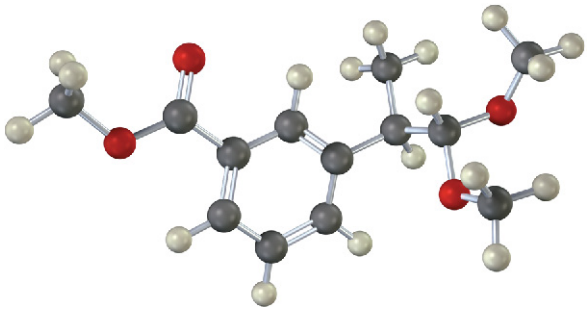
Solution



2-Pentanone

PROBLEM 19-14 Show all the steps in the acid-catalyzed formation of a cyclic acetal from ethylene glycol and an aldehyde or ketone.

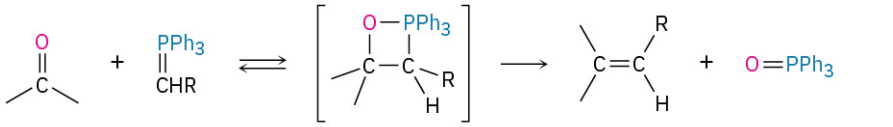
PROBLEM 19-15 Identify the carbonyl compound and the alcohol that were used to prepare the following acetal:



19.11 Nucleophilic Addition of Phosphorus Ylides: The Wittig Reaction

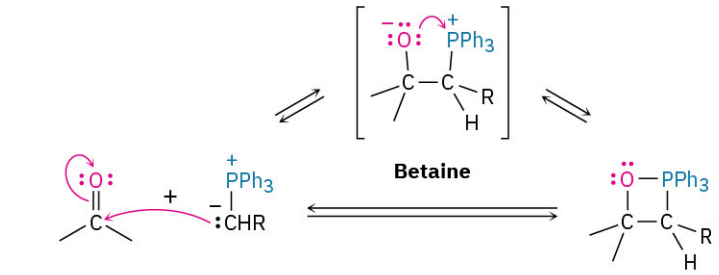
Aldehydes and ketones are converted into alkenes by means of a nucleophilic addition called the **Wittig reaction**, after the German chemist Georg Wittig who won the 1979 Nobel Prize in Chemistry. The reaction has no direct biological counterpart but is important both because of its wide use in drug manufacture and because of its mechanistic similarity to reactions of the coenzyme thiamin diphosphate, which we'll see in **Section 29.6**.

In the Wittig reaction, a triphenylphosphorus *ylide,* $R_2\bar{C}-\overset{+}{P}Ph_3$ also called a **phosphorane** and sometimes written in the resonance form $R_2C{=}P(Ph)_3$, adds to an aldehyde or ketone to yield a four-membered cyclic intermediate called an *oxaphosphetane*. The oxaphosphetane is not isolated, but instead spontaneously decomposes to give an alkene plus triphenylphosphine oxide, $O{=}PPh_3$. In effect, the oxygen atom of the aldehyde or ketone and the $R_2C{=}$ bonded to phosphorus exchange places. (An **ylide**—pronounced **ill**-id—is a neutral, dipolar compound with adjacent positive and negative charges.)

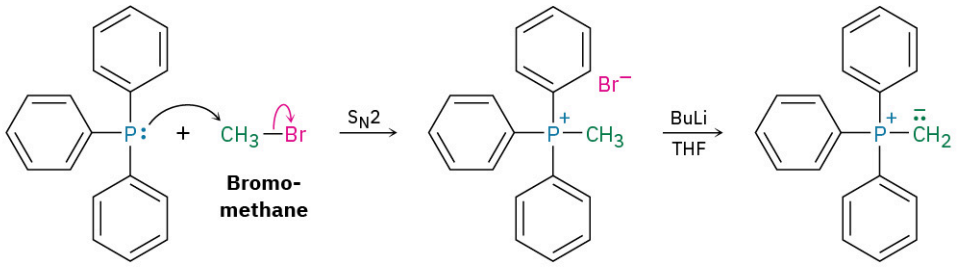


Aldehyde/ketone **Ylide** **Oxaphosphetane** **Alkene** **Triphenylphosphine oxide**

The initial addition takes place by different pathways depending on the structure of the reactants and the exact experimental conditions. One pathway involves a one-step cycloaddition process analogous to the Diels–Alder cycloaddition reaction (**Section 14.4**). The other pathway involves a nucleophilic addition reaction to give a dipolar intermediate called a *betaine* (**bay**-ta-een), which undergoes ring closure.

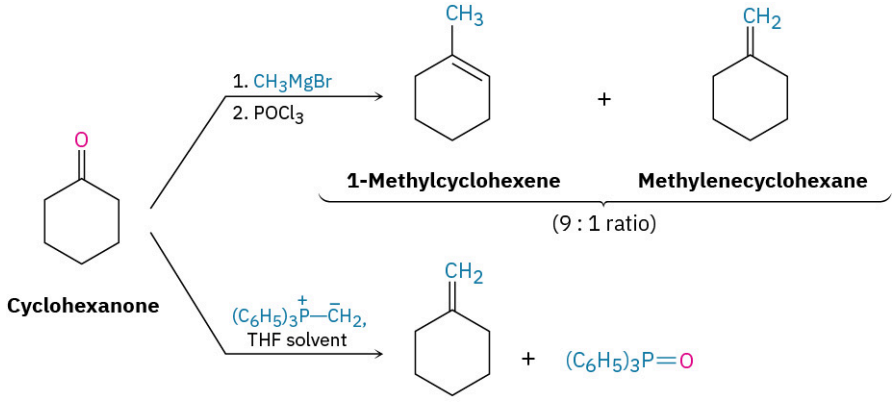


The phosphorus ylides necessary for Wittig reaction are easily prepared by S_N2 reaction of primary (and some secondary) alkyl halides with triphenylphosphine, $(Ph)_3P$, followed by treatment with base. Triphenylphosphine is a good nucleophile in S_N2 reactions, and yields of the resultant alkyltriphenylphosphonium salts are high. Because of the positive charge on phosphorus, the hydrogen on the neighboring carbon is weakly acidic and can be removed by a strong base such as butyllithium (BuLi) to generate the neutral ylide. For example:

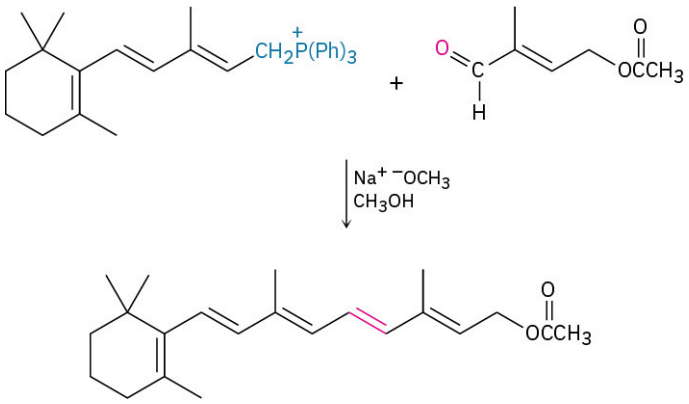


The Wittig reaction is extremely general, and a great many monosubstituted, disubstituted, and trisubstituted alkenes can be prepared from the appropriate combination of phosphorane and aldehyde or ketone. Tetrasubstituted alkenes, however, can't be prepared this way because of steric hindrance.

The real value of the Wittig reaction is that it yields a pure alkene of predictable structure. The C=C bond in the product is always exactly where the C=O group was in the reactant, and no alkene isomers (except *E,Z* isomers) are formed. For example, Wittig reaction of cyclohexanone with methylenetriphenylphosphorane yields only the single alkene product methylenecyclohexane. By contrast, addition of methylmagnesium bromide to cyclohexanone, followed by dehydration with $POCl_3$, yields a roughly 9 : 1 mixture of two alkenes.



Wittig reactions are used commercially in the synthesis of numerous pharmaceutical products. For example, the German chemical company BASF prepares vitamin A by using a Wittig reaction between a 15-carbon ylide and a 5-carbon aldehyde.



Vitamin A acetate

WORKED EXAMPLE 19.3

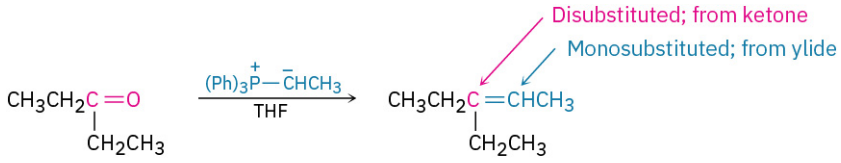
Synthesizing an Alkene Using a Wittig Reaction

What carbonyl compound and what phosphorus ylide might you use to prepare 3-ethyl-2-pentene?

Strategy

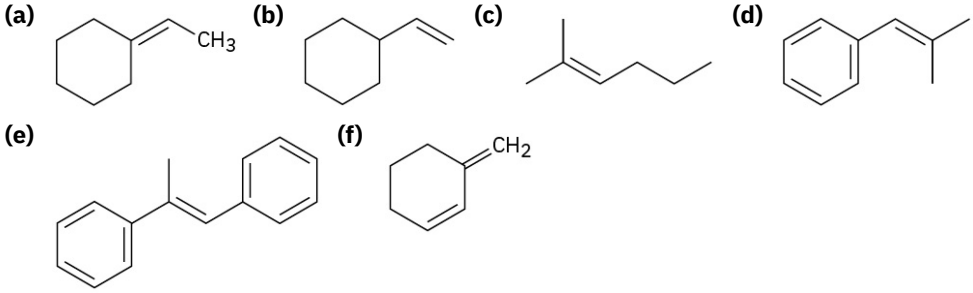
An aldehyde or ketone reacts with a phosphorus ylide to yield an alkene in which the oxygen atom of the carbonyl reactant is replaced by the $=CR_2$ of the ylide. Preparation of the phosphorus ylide itself usually involves S_N2 reaction of a primary alkyl halide with triphenylphosphine, so the ylide is typically primary, $RCH{=}P(Ph)_3$. This means that the disubstituted alkene carbon in the product comes from the carbonyl reactant, while the monosubstituted alkene carbon comes from the ylide.

Solution

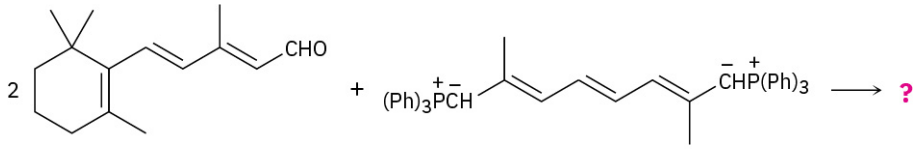


3-Pentanone **3-Ethyl-2-pentene**

PROBLEM 19-16 What carbonyl compound and what phosphorus ylide might you use to prepare each of the following compounds?



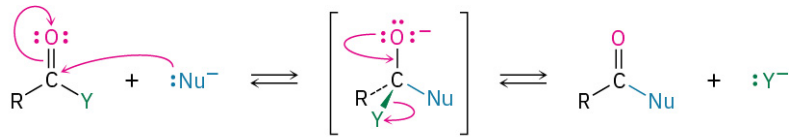
PROBLEM 19-17 *β*-Carotene, a yellow food-coloring agent and dietary source of vitamin A can be prepared by a *double* Wittig reaction between 2 equivalents of *β*-ionylideneacetaldehyde and a *diylide.* Show the structure of the *β*-carotene product.



β-Ionylideneacetaldehyde **A diylide**

19.12 Biological Reductions

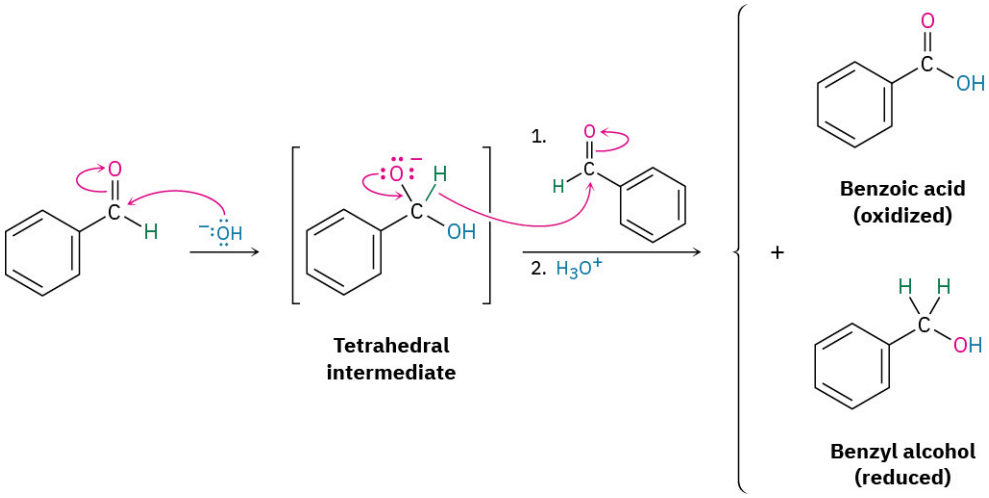
As a general rule, nucleophilic addition reactions are characteristic only of aldehydes and ketones, not of carboxylic acid derivatives. The reason for the difference is structural. As discussed previously in the **Preview of Carbonyl Chemistry**, and shown in **FIGURE 19.12**, the tetrahedral intermediate produced by addition of a nucleophile to a carboxylic acid derivative can eliminate a leaving group, leading to a net nucleophilic acyl substitution reaction. The tetrahedral intermediate produced by addition of a nucleophile to an aldehyde or ketone, however, has only alkyl or hydrogen substituents and thus can't usually expel a leaving group. One exception to this rule is the **Cannizzaro reaction**, discovered in 1853.



Reaction occurs when: Y = –Br, –Cl, –OR, –NR_2
Reaction *does NOT occur* when: Y = –H, –R

FIGURE 19.12 Carboxylic acid derivatives have an electronegative substituent Y = –Br, –Cl, –OR, –NR_2 that can be expelled as a leaving group from the tetrahedral intermediate formed by nucleophilic addition. Aldehydes and ketones have no such leaving group and thus do not usually undergo this reaction.

The Cannizzaro reaction takes place by nucleophilic addition of OH^- to an aldehyde to give a tetrahedral intermediate, which expels hydride ion as a leaving group and is thereby oxidized. A second aldehyde molecule accepts the hydride ion in another nucleophilic addition step and is thereby reduced. Benzaldehyde, for instance, yields benzyl alcohol plus benzoic acid when heated with aqueous NaOH.



The Cannizzaro reaction is rarely used today but is interesting mechanistically because it is a simple laboratory analogy for the primary biological pathway by which carbonyl reductions occur in living organisms. In nature, as we saw in **Section 17.4**, one of the most important reducing agents is NADH, reduced nicotinamide adenine dinucleotide. NADH donates H^- to aldehydes and ketones, thereby reducing them, in much the same way that the tetrahedral alkoxide intermediate in a Cannizzaro reaction does. The electron lone pair on a nitrogen atom of NADH expels H^- as leaving group, which adds to a carbonyl group in another molecule (**FIGURE 19.13**). As an example, pyruvate is converted during intense muscle activity to (*S*)-lactate, a reaction catalyzed by lactate dehydrogenase.

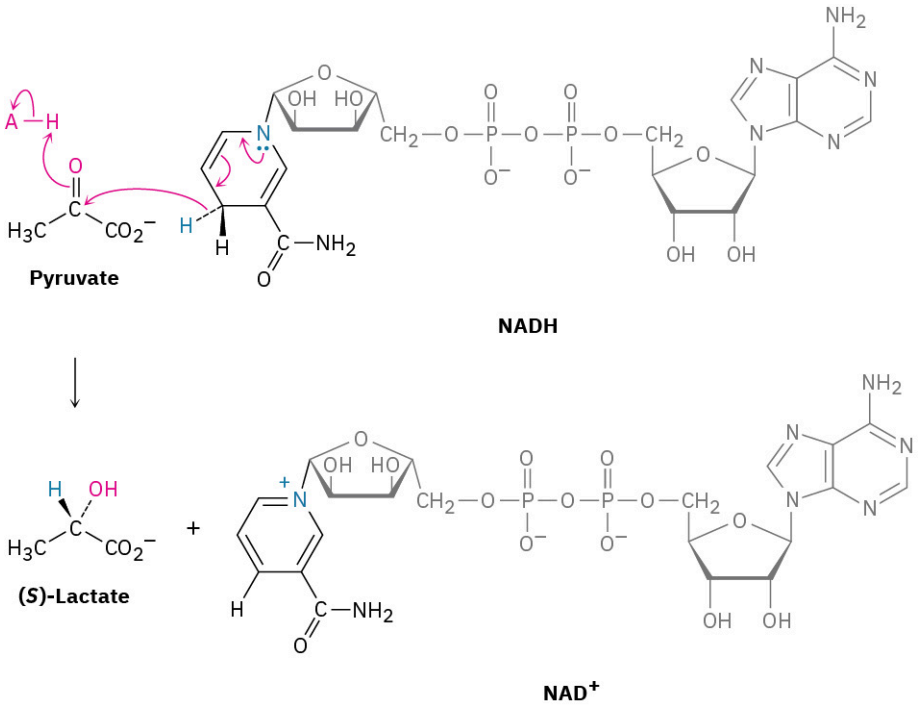
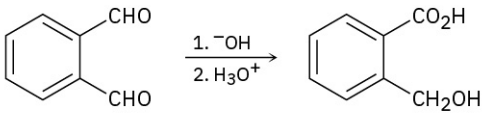


FIGURE 19.13 **Mechanism of biological aldehyde and ketone reductions by the coenzyme NADH.** The key step is an expulsion of hydride ion from NADH and donation to the carbonyl group.

PROBLEM 19-18 When *o*-phthalaldehyde is treated with base, *o*-(hydroxymethyl)benzoic acid is formed. Show the mechanism of this reaction.



***o*-Phthalaldehyde** ***o*-(Hydroxymethyl)benzoic acid**

PROBLEM 19-19 What is the stereochemistry of the pyruvate reduction shown in Figure 19.13? Does NADH lose its pro-*R* or pro-*S* hydrogen? Does addition occur to the Si face or Re face of pyruvate? (Review **Section 5.11**.)

19.13 Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

All the reactions we've been discussing to this point have involved the addition of a nucleophile directly to the carbonyl group, a so-called **1,2-addition**. Closely related to this direct addition is the **conjugate addition**, or **1,4-addition**, of a nucleophile to the C=C bond of an *α*,*β*-unsaturated aldehyde or ketone. (The carbon atom next to a carbonyl group is often called the *α* carbon, the next carbon is the *β* carbon, and so on. Thus, an *α*,*β*-unsaturated aldehyde or ketone has a double bond conjugated with the carbonyl group.) The initial product of conjugate addition is a resonance-stabilized *enolate ion*, which typically undergoes protonation on the *α* carbon to give a saturated aldehyde or ketone product (FIGURE 19.14).

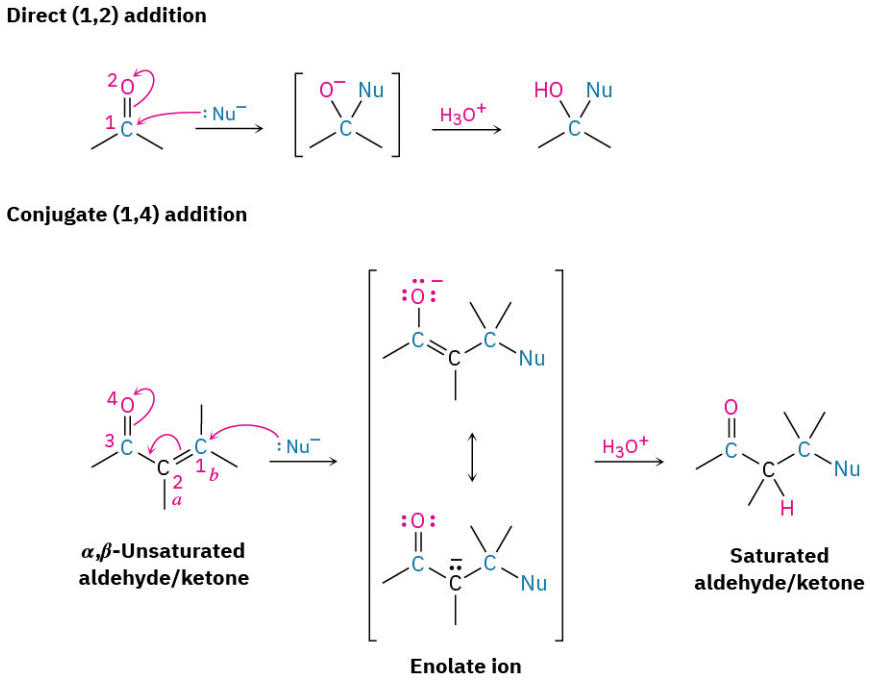
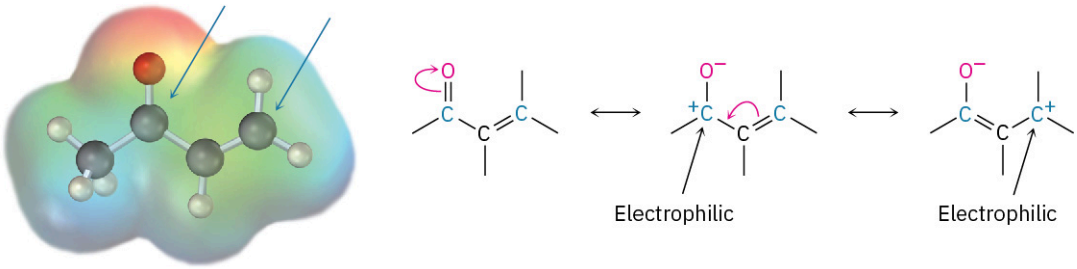
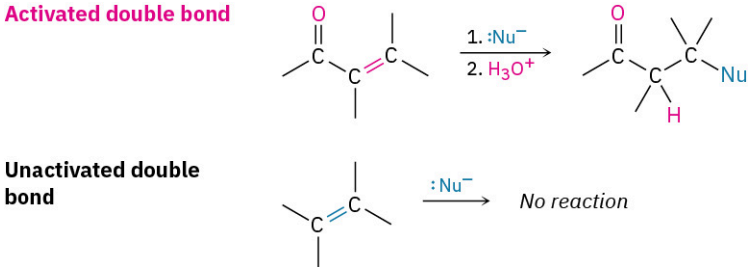


FIGURE 19.14 A comparison of direct (1,2) and conjugate (1,4) nucleophilic addition reactions. In conjugate addition, a nucleophile adds to the β carbon of an α,β-unsaturated aldehyde or ketone and protonation occurs on the α carbon.

The conjugate addition of a nucleophile to an α,β-unsaturated aldehyde or ketone is caused by the same electronic factors that are responsible for direct addition. The electronegative oxygen atom of the α,β-unsaturated carbonyl compound withdraws electrons from the β carbon, thereby making it electron-poor and more electrophilic than a typical alkene carbon.



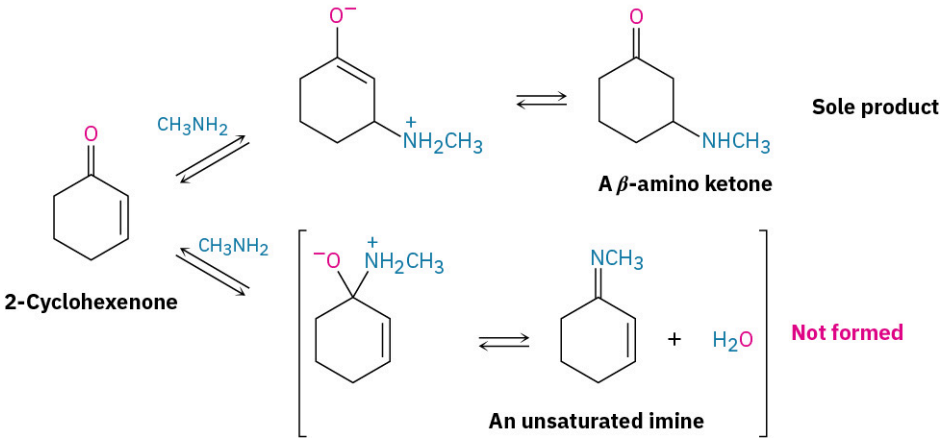
As noted, conjugate addition of a nucleophile to the β carbon of an α,β-unsaturated aldehyde or ketone leads to an enolate ion intermediate, which is protonated on the α carbon to give the saturated product (FIGURE 19.14). The net effect is addition of the nucleophile to the C=C bond, with the carbonyl group itself unchanged. In fact, of course, the carbonyl group is crucial to the success of the reaction. Without the carbonyl group, the C=C bond would not be activated for addition, and no reaction would occur.



Conjugate Addition of Amines

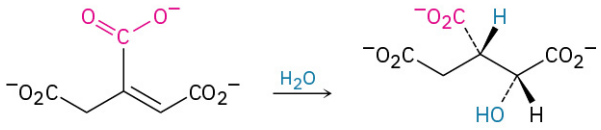
Both primary and secondary amines add to α,β-unsaturated aldehydes and ketones to yield β-amino aldehydes and ketones rather than the alternative imines. Under typical reaction conditions, both modes of addition occur

rapidly. But because the reactions are reversible, they generally proceed with thermodynamic control rather than kinetic control (**Section 14.3**), so the more stable conjugate addition product is often obtained with the complete exclusion of the less stable direct addition product.



Conjugate Addition of Water

Water can add reversibly to α,β-unsaturated aldehydes and ketones to yield β-hydroxy aldehydes and ketones, although the position of the equilibrium generally favors unsaturated reactant rather than saturated adduct. Related additions to α,β-unsaturated carboxylic acids occur in numerous biological pathways, such as the citric acid cycle of food metabolism in which *cis*-aconitate is converted into isocitrate by conjugate addition of water to a double bond.

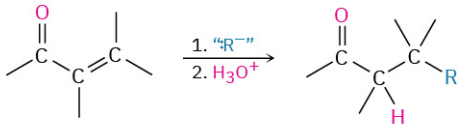


***cis*-Aconitate** **Isocitrate**

PROBLEM 19-20 Assign *R* or *S* stereochemistry to the two chirality centers in isocitrate, and tell whether OH and H add to the Si face or the Re face of the double bond.

Conjugate Addition of Alkyl Groups: Organocopper Reactions

The conjugate addition of an alkyl or other organic group to an α,β-unsaturated ketone (but not aldehyde) is one of the more useful 1,4-addition reactions, just as direct addition of a Grignard reagent is one of the more useful 1,2-additions.



α,β-Unsaturated ketone

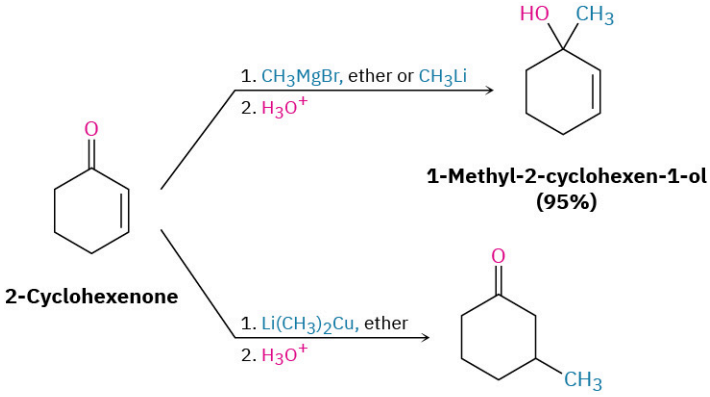
Conjugate addition of an organic group is carried out by treating the α,β-unsaturated ketone with a lithium diorganocopper reagent, R_2CuLi. As we saw in **Section 10.7**, lithium diorganocopper (Gilman) reagents are prepared by reaction between 1 equivalent of copper(I) iodide and 2 equivalents of an organolithium regent, RLi. The organolithium reagent, in turn, is formed by reaction of lithium metal with an organohalide in the same way that a Grignard reagent is prepared by reaction of magnesium metal with an organohalide.

$$RX \xrightarrow[\text{Pentane}]{2\ Li} RLi + Li^+\ X^-$$

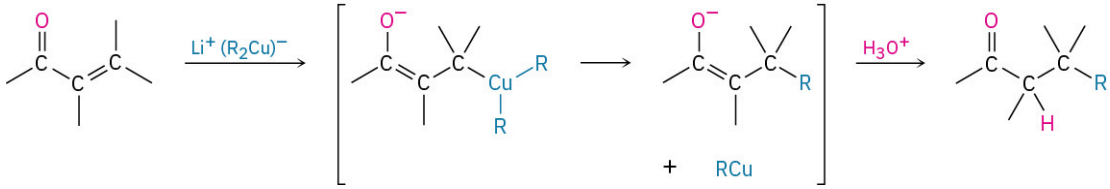
$$2\ RLi \xrightarrow[\text{Ether}]{CuI} Li^+(R\bar{C}uR) + Li^+\ I^-$$

A lithium diorganocopper (Gilman reagent)

Primary, secondary, and even tertiary alkyl groups undergo the conjugate addition reaction, as do aryl and alkenyl groups. Alkynyl groups, however, react poorly in the conjugate addition process. Diorganocopper reagents are unique in their ability to give conjugate addition products. Other organometallic reagents, such as Grignard reagents and organolithiums, normally result in direct carbonyl addition on reaction with α,β-unsaturated ketones.



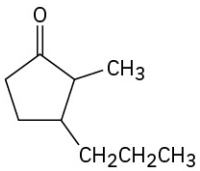
The mechanism of this reaction is thought to involve conjugate nucleophilic addition of the diorganocopper anion, R_2Cu^-, to the unsaturated ketone to give a copper-containing intermediate. Transfer of an R group from copper to carbon, followed by elimination of a neutral organocopper species, RCu, gives the final product.



WORKED EXAMPLE 19.4

Using a Conjugate Addition Reaction

How might you use a conjugate addition reaction to prepare 2-methyl-3-propylcyclopentanone?

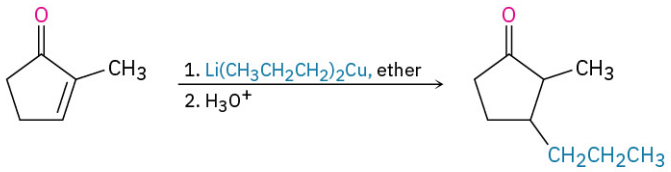


2-Methyl-3-propylcyclopentanone

Strategy

A ketone with a substituent group in its β position might be prepared by a conjugate addition of that group to an α,β-unsaturated ketone. In the present instance, the target molecule has a propyl substituent on the β carbon and might therefore be prepared from 2-methyl-2-cyclopentenone by reaction with lithium dipropylcopper.

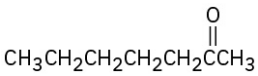
Solution

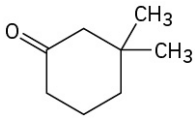


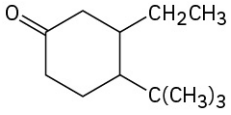
2-Methyl-2-cyclopentenone **2-Methyl-3-propylcyclopentanone**

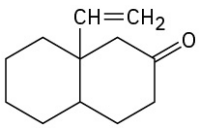
PROBLEM 19-21 Treatment of 2-cyclohexenone with HCN/KCN yields a saturated keto nitrile rather than an unsaturated cyanohydrin. Show the structure of the product, and propose a mechanism for the reaction.

PROBLEM 19-22 How might conjugate addition reactions of lithium diorganocopper reagents be used to synthesize the following compounds?

(a) 

(b) 

(c) 

(d) 

19.14 Spectroscopy of Aldehydes and Ketones

Infrared Spectroscopy

Aldehydes and ketones show a strong C=O bond absorption in the IR region from 1660 to 1770 cm^{-1}, as the spectra of benzaldehyde and cyclohexanone demonstrate (**FIGURE 19.15**). In addition, aldehydes show two characteristic C–H absorptions between 2700–2760 and 2800–2860 cm^{-1}. These absorbances are important for distinguishing between aldehydes and ketones. The higher-frequency absorbance is sometimes obscured in cases where the compound has numerous saturated C–H groups, but the lower-frequency peak is almost always visible.

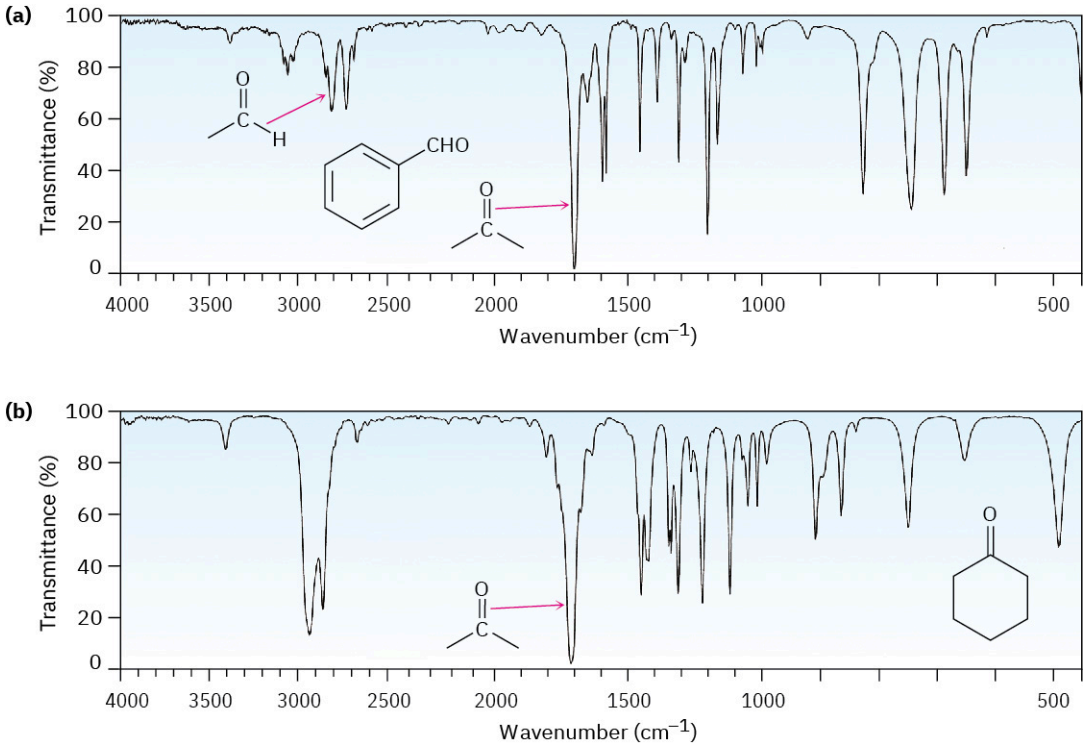


FIGURE 19.15 **Infrared spectra of (a) benzaldehyde and (b) cyclohexanone.**

The exact position of the C=O absorption is diagnostic of the nature of the carbonyl group. As the data in TABLE 19.2 indicate, saturated aldehydes usually show carbonyl absorptions near 1730 cm^{-1} in the IR spectrum, but conjugation of the aldehyde to an aromatic ring or a double bond lowers the absorption by 25 cm^{-1} to near 1705 cm^{-1}. Saturated aliphatic ketones and cyclohexanones both absorb near 1715 cm^{-1}, and conjugation with a double bond or an aromatic ring again lowers the absorption by 30 cm^{-1} to 1685–1690 cm^{-1}. Angle strain in the carbonyl group, caused by reducing the ring size of cyclic ketones to four or five, raises the absorption position. Cyclohexanone has its C=O stretch absorbance at 1715 cm^{-1} while cyclopentanone's carbonyl stretch is at 1750 cm^{-1} and cyclobutanone's is at 1785 cm^{-1}.

TABLE 19.2 **Infrared Absorptions of Some Aldehydes and Ketones**

Carbonyl type	Example	Absorption (cm^{-1})
Saturated aldehyde	CH_3CHO	1730
Aromatic aldehyde	PhCHO	1705
α,β-Unsaturated aldehyde	$H_2C{=}CHCHO$	1705
Saturated ketone	CH_3COCH_3	1715
Cyclohexanone		1715
Cyclopentanone		1750
Cyclobutanone		1785
Aromatic ketone	$PhCOCH_3$	1690
α,β-Unsaturated ketone	$H_2C{=}CHCOCH_3$	1685

The values given in TABLE 19.2 are remarkably constant from one aldehyde or ketone to another. As a result, IR spectroscopy is a powerful tool for identifying the kind of a carbonyl group in a molecule of unknown structure. An unknown that shows an IR absorption at 1730 cm^{-1} is almost certainly an aldehyde rather than a ketone; an unknown that shows an IR absorption at 1750 cm^{-1} is almost certainly a cyclopentanone, and so on.

PROBLEM 19-23 How might you use IR spectroscopy to determine whether reaction between 2-cyclohexenone and lithium dimethylcopper gives the direct addition product or the conjugate addition product?

PROBLEM 19-24 Where would you expect each of the following compounds to absorb in the IR spectrum?
(a) 4-Penten-2-one **(b)** 3-Penten-2-one **(c)** 2,2-Dimethylcyclopentanone
(d) *m*-Chlorobenzaldehyde **(e)** 3-Cyclohexenone **(f)** 2-Hexenal

Nuclear Magnetic Resonance Spectroscopy

Aldehyde protons (RC**H**O) absorb near 10 δ in the ^{1}H NMR spectrum and are very distinctive because no other absorptions occur in this region. The aldehyde proton shows spin–spin coupling with protons on the neighboring carbon, with coupling constant $J \approx 3$ Hz. Acetaldehyde, for example, shows a quartet at 9.79 δ for the aldehyde proton, indicating that there are three protons neighboring the –CHO group (FIGURE 19.16).

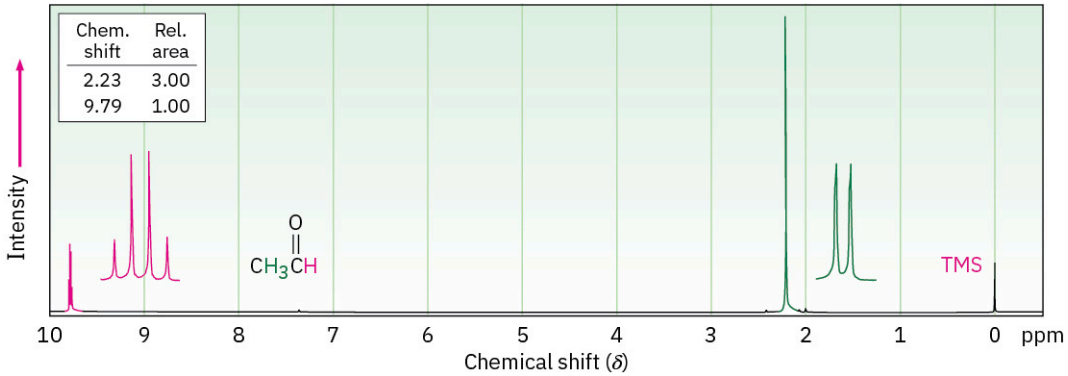
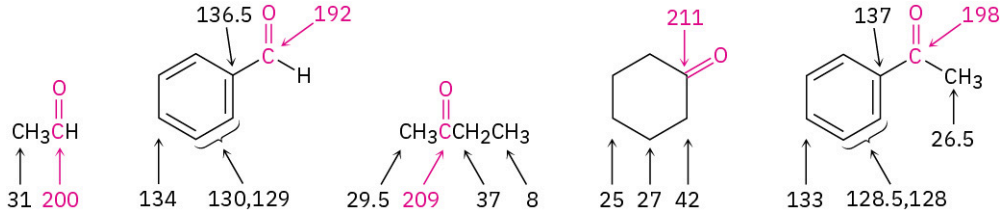


FIGURE 19.16 ^{1}H NMR spectrum of acetaldehyde. The absorption of the aldehyde proton appears at 9.79 δ and is split into a quartet.

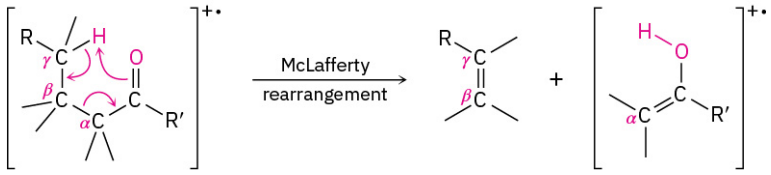
Hydrogens on the carbon next to a carbonyl group are slightly deshielded and usually absorb near 2.0 to 2.3 δ. The acetaldehyde methyl group in FIGURE 19.16, for instance, absorbs at 2.20 δ. Methyl ketones are particularly distinctive because they always show a sharp three-proton singlet near 2.1 δ.

The carbonyl-group carbon atoms of aldehydes and ketones have characteristic ^{13}C NMR resonances in the range 190 to 215 δ. Since no other kinds of carbons absorb in this range, the presence of an NMR absorption near 200 δ is clear evidence for a carbonyl group. Saturated aldehyde or ketone carbons usually absorb in the region from 200 to 215 δ, while aromatic and α,β-unsaturated carbonyl carbons absorb in the 190 to 200 δ region.

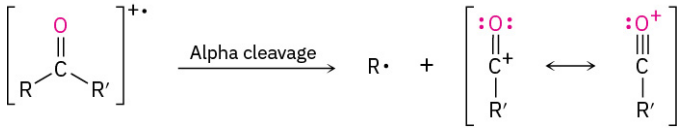


Mass Spectrometry

As discussed in Section 12.3, aliphatic aldehydes and ketones that have hydrogens on their gamma (γ) carbon atoms undergo a characteristic mass spectral cleavage called the **McLafferty rearrangement**. A hydrogen atom is transferred from the γ carbon to the carbonyl oxygen, the bond between the α and β carbons is broken, and a neutral alkene fragment is produced. The charge remains with the oxygen-containing fragment.



In addition to fragmentation by the McLafferty rearrangement, aldehydes and ketones also undergo cleavage of the bond between the carbonyl group and the α carbon, called an α *cleavage*. Alpha cleavage yields a neutral radical and a resonance-stabilized acyl cation.



Fragment ions from both McLafferty rearrangement and α cleavage are visible in the mass spectrum of 5-methyl-2-hexanone shown in **FIGURE 19.17**. McLafferty rearrangement with loss of 2-methylpropene yields a fragment with m/z = 58. Alpha cleavage occurs primarily at the more substituted side of the carbonyl group, leading to a $[CH_3CO]^+$ fragment with m/z = 43.

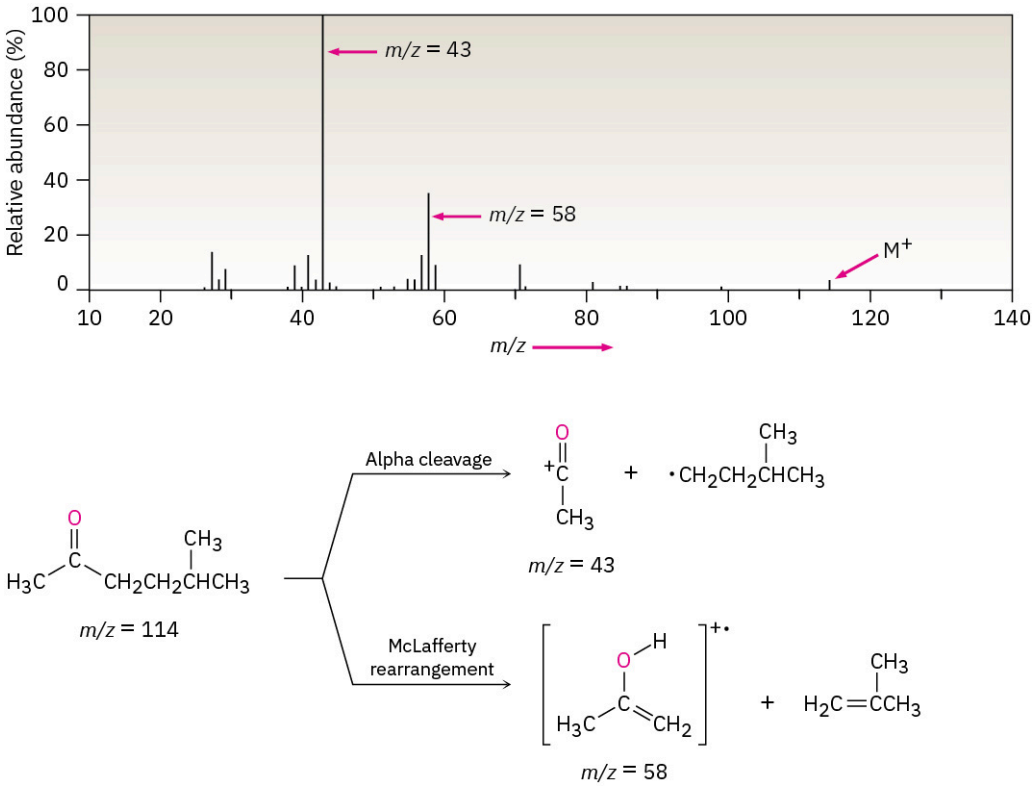
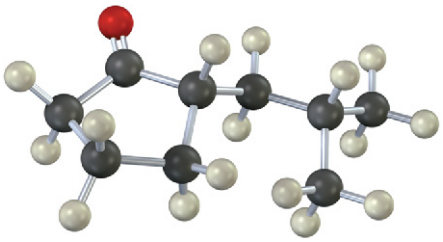


FIGURE 19.17 Mass spectrum and the related reactions of 5-methyl-2-hexanone. The peak at m/z = 58 is due to McLafferty rearrangement. The abundant peak at m/z = 43 is due to α cleavage at the more highly substituted side of the carbonyl group. Note that the peak due to the molecular ion is very small.

PROBLEM 19-25 How might you use mass spectrometry to distinguish between the following pairs of isomers?
(a) 3-Methyl-2-hexanone and 4-methyl-2-hexanone **(b)** 3-Heptanone and 4-heptanone
(c) 2-Methylpentanal and 3-methylpentanal

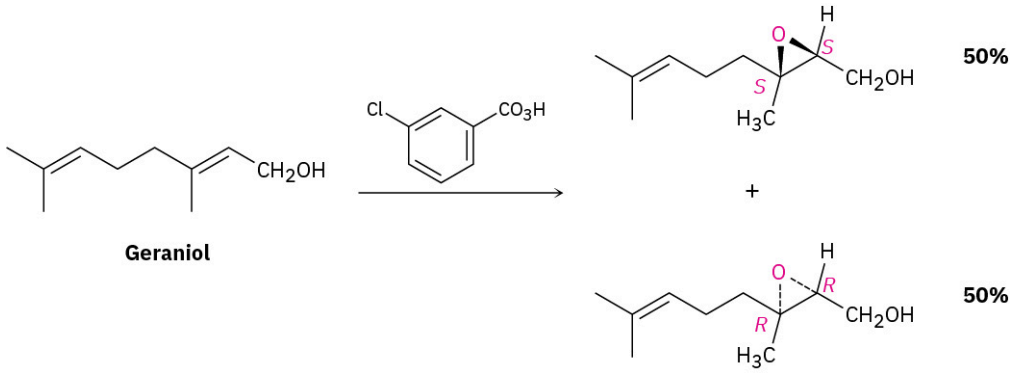
PROBLEM 19-26 Describe the prominent IR absorptions and mass spectral peaks you would expect for the following compound:



CHEMISTRY MATTERS

Enantioselective Synthesis

Whenever a chiral product is formed by reaction between achiral reagents, the product is racemic; that is, both enantiomers of the product are formed in equal amounts. The epoxidation reaction of geraniol with *m*-chloroperoxybenzoic acid, for instance, gives a racemic mixture of (2*S*,3*S*) and (2*R*,3*R*) epoxides.



Unfortunately, it's usually the case that only one enantiomer of a given drug or other important substance has the desired biological properties. The other enantiomer might be inactive or even dangerous. Thus, much work is currently being done on developing *enantioselective* methods of synthesis, which yield only one of the two possible enantiomers. So important has enantioselective synthesis become that the 2001 Nobel Prize in Chemistry was awarded to three pioneers in the field: William S. Knowles of the Monsanto Co., K. Barry Sharpless of the Scripps Research Institute, and Ryoji Noyori of Nagoya University.

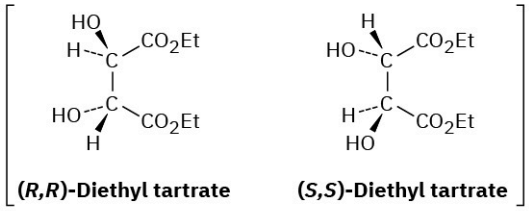
Several approaches to enantioselective synthesis have been developed, but the most efficient are those that use chiral catalysts to temporarily hold a substrate molecule in an unsymmetrical environment—the same strategy that nature uses when catalyzing reactions with chiral enzymes. While in that unsymmetrical environment, the substrate may be more open to reaction on one side than on another, leading to an excess of one enantiomeric product over another. As an analogy, think about picking up a coffee mug in your right hand to take a drink. The mug by itself is achiral, but as soon as you pick it up by the handle, it becomes chiral. One side of the mug now faces toward you so you can drink from it, but the other side faces away. The two sides are different, with one side much more accessible to you than the other.

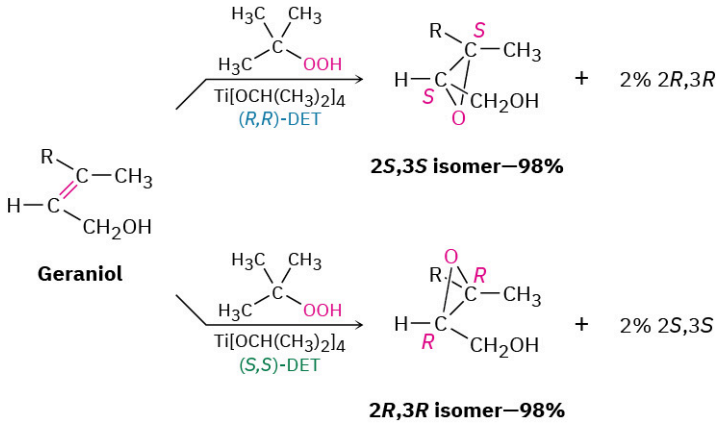


FIGURE 19.18 A substance made from the tartaric acid found at the bottom of this wine vat catalyzes enantioselective reactions. (credit: "Pinot Noir fermentation in open vat" by Pipers Brook Vineyard Media/Flickr, CC BY 2.0)

Among the thousands of enantioselective reactions now known, one of the most useful is the so-called Sharpless epoxidation, in which an allylic alcohol, such as geraniol, is treated with *tert*-butyl hydroperoxide,

$(CH_3)_3C–OOH$, in the presence of titanium tetraisopropoxide and diethyl tartrate (DET) as a chiral auxiliary reagent. When (*R,R*) tartrate is used, geraniol is converted into its 2*S*,3*S* epoxide with 98% selectivity, whereas the use of (*S,S*) tartrate gives the 2*R*,3*R* epoxide enantiomer. We say that the major product in each case is formed with an *enantiomeric excess* of 96%, meaning that 4% of the product is racemic (2% 2*S*,3*S* plus 2% 2*R*,3*R*) and an extra 96% of a single enantiomer is formed. The mechanistic details by which the chiral catalyst works are a bit complex, although it appears that a chiral complex of two tartrate molecules with one titanium is involved.





Key Terms

- **acetal, $R_2C(OR')_2$**
- **acyl group**
- **1,2-addition**
- **1,4-addition**
- **aldehyde (RCHO)**
- **Cannizzaro reaction**
- **carbinolamine**
- **conjugate addition**
- **cyanohydrin**
- **enamine** ($R_2N–CR{=}CR_2$)
- **hemiacetal**
- **imine** ($R_2C{=}NR$)
- **ketal**
- **ketone (R_2CO)**
- **McLafferty rearrangement**
- **nucleophilic addition reaction**
- **phosphorane**
- **Schiff bases**
- **Wittig reaction**
- **Wolff–Kishner reaction**
- **ylide**

Summary

Aldehydes and ketones are among the most important of all functional groups, both in the chemical industry and in biological pathways. In this chapter, we've looked at some of their typical reactions. Aldehydes are normally prepared in the laboratory by oxidation of primary alcohols or by partial reduction of esters. Ketones are similarly prepared by oxidation of secondary alcohols.

The **nucleophilic addition reaction** is the most common general reaction type for aldehydes and ketones. Many different kinds of products can be prepared by nucleophilic additions. Aldehydes and ketones are reduced by $NaBH_4$ or $LiAlH_4$ to yield primary and secondary alcohols, respectively. Addition of Grignard reagents to aldehydes and ketones also gives alcohols (secondary and tertiary, respectively), and addition of HCN yields **cyanohydrins**. Primary amines add to carbonyl compounds yielding **imines**, or **Schiff bases**, and secondary

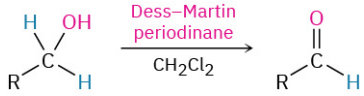
amines yield **enamines**. Reaction of an aldehyde or ketone with hydrazine and base gives an alkane (the **Wolff–Kishner reaction**). Alcohols add to carbonyl groups to yield **acetals**, which are valuable as protecting groups. Phosphorus **ylides** add to aldehydes and ketones in the **Wittig reaction** to give alkenes.

α,β-Unsaturated aldehydes and ketones often react with nucleophiles to give the product of **conjugate addition**, or **1,4-addition**. Particularly useful are the conjugate addition of an amine and the conjugate addition of an organic group by reaction with a diorganocopper reagent.

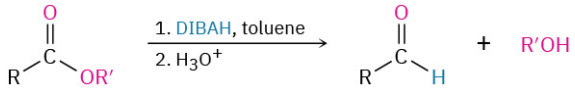
IR spectroscopy is helpful for identifying aldehydes and ketones. Carbonyl groups absorb in the IR range 1660 to 1770 cm^{-1}, with the exact position highly diagnostic of the kind of carbonyl group present in the molecule. ^{13}C NMR spectroscopy is also useful for aldehydes and ketones because their carbonyl carbons show resonances in the 190 to 215 δ range. ^{1}H NMR is useful for aldehyde –CHO protons, which absorb near 10 δ. Aldehydes and ketones undergo two characteristic kinds of fragmentation in the mass spectrometer: α cleavage and McLafferty rearrangement.

Summary of Reactions

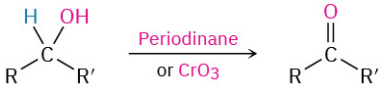
1. Preparation of aldehydes (Section 19.2)
   a. Oxidation of primary alcohols (Section 17.7)



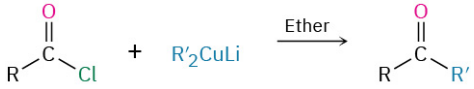
   b. Partial reduction of esters (Section 19.2)



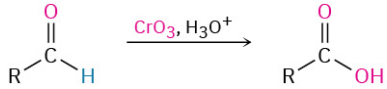
2. Preparation of ketones
   a. Oxidation of secondary alcohols (Section 17.7)



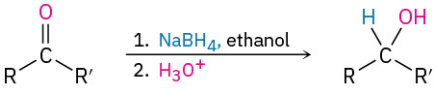
   b. Diorganocopper reaction with acid chlorides (Section 19.2)



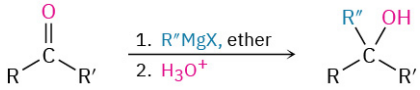
3. Oxidation of aldehydes (Section 19.3)



4. Nucleophilic addition reactions of aldehydes and ketones
   a. Addition of hydride to give alcohols: reduction (Section 19.7)



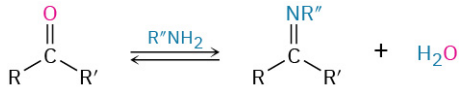
   b. Addition of Grignard reagents to give alcohols (Section 19.7)



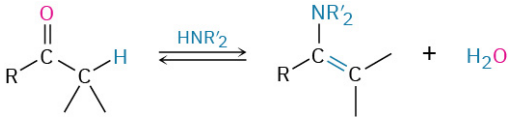
   c. Addition of HCN to give cyanohydrins (Section 19.6)



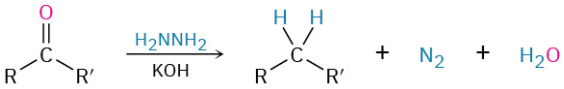
   d. Addition of primary amines to give imines (Section 19.8)



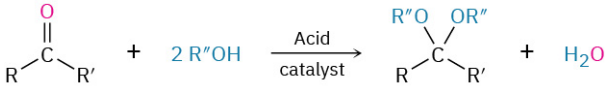
e. Addition of secondary amines to give enamines (**Section 19.8**)



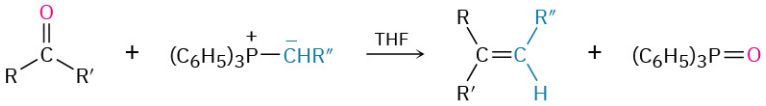
f. Wolff–Kishner reaction to give alkanes (**Section 19.9**)



g. Addition of alcohols to give acetals (**Section 19.10**)

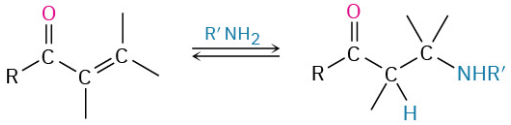


h. Addition of phosphorus ylides to give alkenes: Wittig reaction (**Section 19.11**)

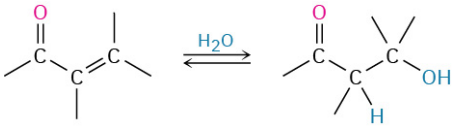


5. Conjugate additions to α,β-unsaturated aldehydes and ketones (**Section 19.13**)

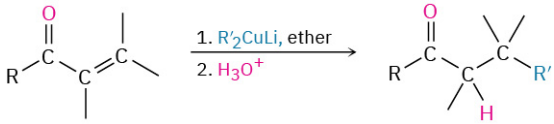
a. Conjugate addition of amines



b. Conjugate addition of water



c. Conjugate addition of alkyl groups by diorganocopper reaction

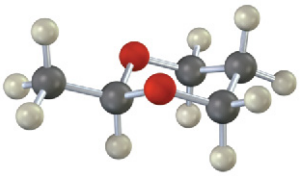


Additional Problems

Visualizing Chemistry

PROBLEM 19-27 Each of the following substances can be prepared by a nucleophilic addition reaction between an aldehyde or ketone and a nucleophile. Identify the reactants from which each was prepared. If the substance is an acetal, identify the carbonyl compound and the alcohol; if it is an imine, identify the carbonyl compound and the amine; and so forth.

(a)



(b)

